

Self-contact frictional jamming of helical ribbons

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September 9, 2026

Abstract

A paper yoyo is a ribbon wound around itself in self-contact and pulled apart along its winding axis. We derive a quasi-static continuum model of the resulting force-displacement relation $F(x)$ by deriving and assembling five independently derived components: the winding kinematics, the three-dimensional stress balance, constitutive relations and compatibility, Coulomb friction at the self-contact interfaces, and elastic resistance to bending away from the ribbon’s natural curl. The assembly produces two coupled terms of $F(x)$. First is the conjugate force of the stored bending energy, which is fully determined by the ribbon’s geometry and elastic moduli. The second is the friction-driven membrane stress, which requires a normalization that the friction equations cannot supply on their own; we show that the ribbon’s own arc-length constraint supplies it, and solve the resulting coupled system numerically. Along the way we identify a “locking” phenomena at which it becomes more energetically favorable to stretch the material rather than kinematically tighten the coil.

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1 The system, and the assumptions

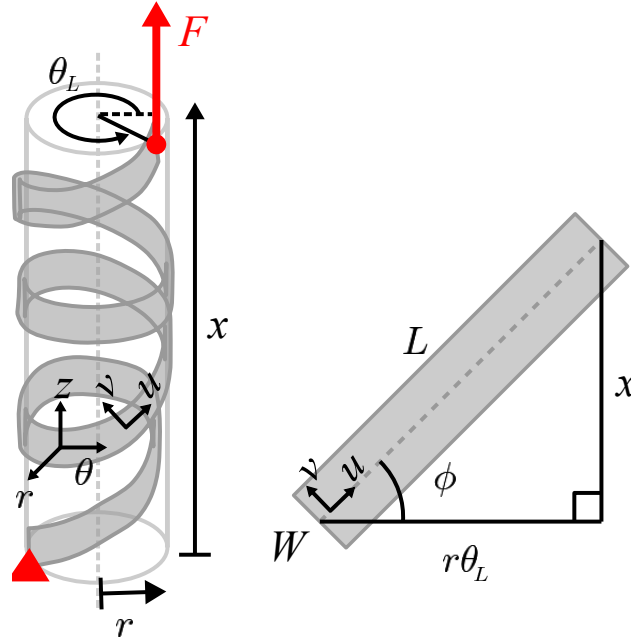


Figure 1: The wound ribbon. One end is held fixed; the other (“active”) end is displaced by x along the winding axis.

The system is a long constant-width rectangular ribbon of small but finite thickness, wrapped around itself at a uniform pitch. One end of the ribbon is fixed to prevent rigid-body motion; the other is displaced by a distance x along the winding axis, which is taken to be the z -axis (Figure 1).

1.1 Geometric and material constants

The geometric and material constants are

- L : the length of the ribbon in its natural, unstressed state.
- W : its width, with $W \ll L$.
- t : its thickness, with $t \ll W$.
- θ_L : the total wind angle. This is *fixed* by the boundary conditions: the two ends cannot rotate about z relative to one another, only translate along it, so the number of turns is a topological property of how the ribbon was initially wound and does not change as it is pulled. $\theta_L = 2\pi n$ where n is the (not necessarily integer) number of turns.
- E, ν : Young's modulus and Poisson ratio of the ribbon's material.
- μ : the Coulomb friction coefficient for sheet-on-sheet contact.

1.2 Assumptions

It is worth putting these in one place, since several of the model's conclusions turn on them.

1. **Quasi-static loading.** Inertia is neglected; the state at each x is an equilibrium state.
2. **Uniform pitch angle ϕ .** The angle the local ribbon direction makes with the (r, θ) plane is the same everywhere on the sheet at a given x . This is a geometric simplification that keeps the surface developable (Section 2.3).
3. **Uniform coil radius r .** At a given x , every material point sits at the same radius from the winding axis. This is another geometric simplification: successive turns really differ in radius by one thickness, so r varies by $t\theta_L/2\pi$ across the ribbon. The fractional variation is $t\theta_L/(2\pi r)$, small for a coil wound with a central hole $r \gg t$. We retain the effect where it matters—it sets the packing floor of Section 2.4—but not in the stress field.
4. **Linear elasticity, small strain, and plane stress.** The sheet is Hookean, and thin enough that the through-thickness stress state is plane.
5. **Linear variation of every stress component through the thickness.** Because t is small, the variation of values across the thickness can be approximated as linear. Therefore, knowing a component's two surface values is then equivalent to knowing its full profile by linear interpolation.
6. **Coulomb friction with a velocity-aligned direction.** At each contact point between successive layers, the shear traction has magnitude μ times the normal pressure and opposes the local relative sliding velocity. This is a sliding (kinetic) law, so it presumes the interface is slipping rather than stuck. Section 5.2 shows the relative slip velocity is nonzero for every $x > 0$, so the presumption is self-consistent, with the single exception of $x = 0$ exactly.
7. **Contact transmits pressure and friction only, with no adhesion.** Two surfaces can push on each other but not pull, so $\sigma_{rr} \leq 0$. We state this as a modelling requirement but do *not* enforce it as an inequality constraint; Section 9.3 reports where the solution violates it.

8. **Kirchhoff normality.** Material lines initially normal to the mid-surface remain normal to it and remain unstretched, so the transverse shear strains vanish, $\epsilon_{ur} = \epsilon_{vr} = 0$. This is the standard thin-plate kinematic hypothesis, admissible because $t \ll W$, and it is what makes the moment closure of Section 6.2 meaningful. Its consequences are worked out in Section 11.2; the most important is that σ_{ur} and σ_{vr} become workless constraint reactions.
9. **Slender through-thickness measure.** Wherever a volume or face element is integrated across the thickness, the curvature weights of Eq. (115) are set to unity, i.e. $|\zeta| \leq t/2$ is treated as small against r . Section 11.5 bounds the neglected terms and names the one energy contribution they carry. This is the same approximation already made in Section 3.2, where the faces of a volume element are treated as rectangular.
10. **Saint-Venant interior.** In the friction-free problem of Section 11 we work with the interior solution, discarding two boundary layers: one of extent $\sim W$ at each gripped end, and one of extent $\sim \sqrt{rt}$ along each long edge. This is not an assumption about uniformity but about geometry, and it is charged to $W \ll L$ and $t \ll W$ from Section 1.1: within the interior, Section 11.8 *derives* that every $\frac{\partial}{\partial u}$ and $\frac{\partial}{\partial v}$ of an in-plane stress vanishes, by solving the $\hat{u}$ and $\hat{v}$ equilibrium equations rather than by dropping them. Two limits are worth recording here. Section 11.7 shows the derivation fails under sliding friction, which supplies a distributed in-plane load and so has no uniform interior — that case belongs to Section 8. And Section 11.9 identifies the one equation the interior solution cannot satisfy.

2 Kinematics

This section derives the system geometry before introducing forces.

2.1 Variables

The single defining state variable is x . All other variables describe the system at one frozen value of x , or about how it changes as x varies.

The key variables to describe the geometry of a given state are the coil radius r and the pitch angle ϕ . As mentioned in Section 1.2, we assume both are uniform across the ribbon at a given x . The only other geometric variable is the arc length of a material line along the ribbon (i.e., the length of the ribbon), which we denote Λ . It is related to r and x by the Pythagorean relation

$$\Lambda = \sqrt{(r \theta_L)^2 + x^2}. \quad (1)$$

If the ribbon carries a mean axial strain $\epsilon := \epsilon_{uu}$, that same material line has length $L(1 + \epsilon)$, so

$$\Lambda = L(1 + \epsilon), \quad r \theta_L = \sqrt{L^2(1 + \epsilon)^2 - x^2}. \quad (2)$$

By unrolling this strained ribbon into a the hypotenuse of a right triangle, we can relate

$$\sin \phi = \frac{x}{\Lambda} \quad \text{and} \quad \cos \phi = \frac{r \theta_L}{\Lambda} \quad (3)$$

Then,

$$x = L(1 + \epsilon) \sin \phi \quad \xrightarrow{\epsilon \rightarrow 0} \quad x = L \sin \phi, \quad r \theta_L = L \cos \phi. \quad (4)$$

Observe that for a fixed ϵ , increasing x forces ϕ to increase, which causes r to decrease. In other words, the coil tightens as it is pulled. It is worth being explicit that this tightening is a purely kinematic rearrangement which costs no stretching: Section 2.3 shows the deformation is an exact isometry.

We now define the following nondimensional variables:

$$\boxed{r_{\text{nat}} := \frac{L}{\theta_L}, \quad \rho := \frac{r}{r_{\text{nat}}}, \quad \chi := \frac{x}{L}, \quad \lambda := \frac{\Lambda}{L} = \sqrt{\rho^2 + \chi^2} = 1 + \epsilon.} \quad (5)$$

- r_{nat} is the coil radius at the resting state at $x = 0$, or in other words, the radius of the material's natural curvature. Changing the curvature of the material produces bending stress. It has units of length.
- ρ is the current coil radius normalized by r_{nat} .
- χ is the current axial displacement normalized by the ribbon length.
- λ is the current arc length normalized by the ribbon length.

2.2 Position vector and local frame

The position of a point on the ribbon can be described either by surface coordinates (u, v) on the unrolled rectangle— $u \in [0, L]$ along its length, $v \in [-W/2, W/2]$ across its width—or by cylindrical coordinates (r, θ, z) in space. The through-thickness coordinate is $\zeta \in [-t/2, t/2]$, measured from the mid-surface along the outward radial direction $\hat{r}$, so a general material point sits at $\vec{p}(u, v, x) + \zeta \hat{r}$.

Uniform pitch angle ϕ means the local ribbon direction is inclined at ϕ to the (r, θ) plane, so

$$\frac{\partial z}{\partial u} = \sin \phi, \quad \frac{\partial z}{\partial v} = \cos \phi, \quad (6)$$

and, at uniform radius, the circumferential relations are

$$r \frac{\partial \theta}{\partial u} = \cos \phi, \quad r \frac{\partial \theta}{\partial v} = -\sin \phi. \quad (7)$$

With r uniform, Eqs. (6)–(7) integrate directly to obtain

$$\boxed{\vec{p}(u, v, x) = \begin{bmatrix} r \\ \theta \\ z \end{bmatrix} = \begin{bmatrix} r(x) \\ \frac{u \cos \phi - v \sin \phi}{r(x)} \\ u \sin \phi + v \cos \phi \end{bmatrix}}. \quad (8)$$

Differentiating in the cylindrical basis, using $\frac{\partial \hat{r}}{\partial \theta} = \hat{\theta}$, $\frac{\partial \hat{\theta}}{\partial \theta} = -\hat{r}$ and $\frac{\partial \hat{z}}{\partial \theta} = 0$,

$$\frac{\partial \vec{p}}{\partial u} = \cos \phi \hat{\theta} + \sin \phi \hat{z}, \quad \frac{\partial \vec{p}}{\partial v} = -\sin \phi \hat{\theta} + \cos \phi \hat{z}, \quad (9)$$

which are exactly orthonormal, so we may set $\hat{u} := \frac{\partial \vec{p}}{\partial u}$ and $\hat{v} := \frac{\partial \vec{p}}{\partial v}$ with no approximation.

2.3 Curvature and developability

Differentiating $\hat{u}$ and $\hat{v}$ once more and reading off the components along $\hat{r}$ gives the second fundamental form:

$$\boxed{\kappa_{uu} = \frac{\cos^2 \phi}{r}, \quad \kappa_{uv} = \frac{\sin \phi \cos \phi}{r}, \quad \kappa_{vv} = \frac{\sin^2 \phi}{r}}, \quad (10)$$

Together with the frame-rotation rates needed in Section 3, the Gaussian curvature vanishes:

$$\frac{\partial \hat{r}}{\partial u} = \kappa_{uu} \hat{u} - \kappa_{uv} \hat{v}, \quad \frac{\partial \hat{r}}{\partial v} = -\kappa_{uv} \hat{u} + \kappa_{vv} \hat{v}. \quad (11)$$

$$\boxed{K = \kappa_{uu}\kappa_{vv} - \kappa_{uv}^2 = \frac{\sin^2 \phi \cos^2 \phi}{r^2} - \frac{\sin^2 \phi \cos^2 \phi}{r^2} = 0}. \quad (12)$$

Therefore, the surface is always developable.

2.4 The minimum radius

The one place the finite thickness of the stack must be retained is the limit on how far the coil can tighten. Successive turns are separated radially by one thickness, so over θ_L of winding the coil occupies a radial band of extent $t\theta_L/2\pi$. The mean radius cannot fall below half that band without the innermost turn closing on the axis (that is, radius exactly zero), which defines the scale

$$\boxed{\rho_{\text{jam}} = \frac{t\theta_L^2}{4\pi L} = \frac{tn}{2r_{\text{nat}}}, \quad n = \frac{\theta_L}{2\pi}}. \quad (13)$$

The second form reads transparently: ρ_{jam} is half the coil's total radial build measured against its initial radius. Section 7.3 shows that this is an asymptote and not a barrier: the bending energy of the innermost turn diverges as its radius goes to zero, so ρ_{jam} is approached and never reached. A coil wound with a large central hole would have $tn \ll r_{\text{nat}}$ and $\rho_{\text{jam}} \ll 1$; one with a tight initial winding would have $tn \rightarrow r_{\text{nat}}$ and $\rho_{\text{jam}} \rightarrow 1/2$.

3 Stress balance

3.1 Notation

At a point (u, v, ζ) the material carries the full Cauchy stress

$$\sigma = \begin{pmatrix} \sigma_{uu} & \sigma_{uv} & \sigma_{ur} \\ \sigma_{uv} & \sigma_{vv} & \sigma_{vr} \\ \sigma_{ur} & \sigma_{vr} & \sigma_{rr} \end{pmatrix}, \quad (14)$$

with six independent components: the in-plane stresses $\sigma_{uu}, \sigma_{uv}, \sigma_{vv}$; the transverse shears σ_{ur}, σ_{vr} , by which the ribbon transmits load sideways through its thickness; and σ_{rr} , the contact pressure between layers.

Two kinds of through-thickness reduction recur. Thickness averages are written with a bar. Under the linear-profile assumption (assumption 5, Section 1.2) the average of a component equals the mean of its two surface values:

$$\bar{\sigma}_{ij} := \frac{1}{t} \int_{-t/2}^{t/2} \sigma_{ij} d\zeta = \frac{1}{2} \left[\sigma_{ij} \left(\zeta = \frac{t}{2} \right) + \sigma_{ij} \left(\zeta = -\frac{t}{2} \right) \right]. \quad (15)$$

We also define the difference of the two surface values using a Δ :

$$\Delta\sigma_{ij} := \sigma_{ij} \left(\zeta = \frac{t}{2} \right) - \sigma_{ij} \left(\zeta = -\frac{t}{2} \right). \quad (16)$$

Finally, the two surface values of the contact pressure σ_{rr} get their own names:

$$\sigma_{rr}^{\pm} := \sigma_{rr}(\zeta = \pm \frac{t}{2}) \quad (17)$$

3.2 Force balance on a curved element

Perform a force balance on a volume element $du \times dv \times d\zeta$. On its u -face the traction is $\vec{t}_u = \sigma_{uu}\hat{u} + \sigma_{uv}\hat{v} + \sigma_{ur}\hat{r}$, and similarly $\vec{t}_v$ and $\vec{t}_\zeta$ on the other two. The faces of the element are approximately rectangular because $t \ll r$, so

$$\left(\frac{\partial \vec{t}_u}{\partial u} + \frac{\partial \vec{t}_v}{\partial v} + \frac{\partial \vec{t}_\zeta}{\partial \zeta} \right) du dv d\zeta = 0. \quad (18)$$

The basis vectors $\hat{u}$, $\hat{v}$, and $\hat{r}$ themselves depend on position (Eq. 11), so differentiating the tractions produces curvature-weighted terms beyond the plain derivatives of the stress components. For example,

$$\frac{\partial}{\partial u} (\sigma_{uu}\hat{u}) = \frac{\partial \sigma_{uu}}{\partial u} \hat{u} + \sigma_{uu} \frac{\partial \hat{u}}{\partial u} = \frac{\partial \sigma_{uu}}{\partial u} \hat{u} - \kappa_{uu} \sigma_{uu} \hat{r}, \quad (19)$$

The extra term expresses the fact that the element is not planar, so in-plane stresses are slightly rotated into the out-of-plane direction, and vice versa. Expanding every term this way and sorting

by component gives three scalar equations:

$$\hat{u} : \quad \frac{\partial \sigma_{uu}}{\partial u} + \frac{\partial \sigma_{uv}}{\partial v} + \frac{\partial \sigma_{ur}}{\partial \zeta} + \kappa_{uu} \sigma_{ur} - \kappa_{uv} \sigma_{vr} = 0 \quad (20)$$

$$\hat{v} : \quad \frac{\partial \sigma_{uv}}{\partial u} + \frac{\partial \sigma_{vv}}{\partial v} + \frac{\partial \sigma_{vr}}{\partial \zeta} - \kappa_{uv} \sigma_{ur} + \kappa_{vv} \sigma_{vr} = 0 \quad (21)$$

$$\hat{r} : \quad \frac{\partial \sigma_{ur}}{\partial u} + \frac{\partial \sigma_{vr}}{\partial v} + \frac{\partial \sigma_{rr}}{\partial \zeta} - \kappa_{uu} \sigma_{uu} + 2\kappa_{uv} \sigma_{uv} - \kappa_{vv} \sigma_{vv} = 0 \quad (22)$$

Equilibrium alone provides three equations for these six unknown stress components. The remaining relations come from friction (Section 5), which supplies the surface values of σ_{ur} , σ_{vr} , σ_{rr} , and from compatibility (Section 4).

3.3 Zeroth thickness moment

Averaging Eqs. (20)–(22) over the thickness, and using $\frac{1}{t} \int \frac{\partial \sigma_{ur}}{\partial \zeta} d\zeta = \frac{1}{t} \Delta \sigma_{ur}$,

$$\frac{\partial \bar{\sigma}_{uu}}{\partial u} + \frac{\partial \bar{\sigma}_{uv}}{\partial v} + \frac{1}{t} \Delta \sigma_{ur} + \kappa_{uu} \bar{\sigma}_{ur} - \kappa_{uv} \bar{\sigma}_{vr} = 0 \quad (23)$$

$$\frac{\partial \bar{\sigma}_{uv}}{\partial u} + \frac{\partial \bar{\sigma}_{vv}}{\partial v} + \frac{1}{t} \Delta \sigma_{vr} - \kappa_{uv} \bar{\sigma}_{ur} + \kappa_{vv} \bar{\sigma}_{vr} = 0 \quad (24)$$

$$\frac{\partial \bar{\sigma}_{ur}}{\partial u} + \frac{\partial \bar{\sigma}_{vr}}{\partial v} + \frac{1}{t} \Delta \sigma_{rr} - \kappa_{uu} \bar{\sigma}_{uu} + 2\kappa_{uv} \bar{\sigma}_{uv} - \kappa_{vv} \bar{\sigma}_{vv} = 0 \quad (25)$$

These are still exact consequences of Eqs. (20)–(22); no term has been dropped.

3.4 First order moment balance

Multiplying by ζ and integrating Eqs. (20)–(22) gives an independent statement—the moment balance—which Section 6.3 needs. Define the bending moments and the first moment of the transverse shear,

$$M_{ij} := \int_{-t/2}^{t/2} \zeta \sigma_{ij} d\zeta \quad (ij \in \{uu, uv, vv\}), \quad \Xi_{ir} := \int_{-t/2}^{t/2} \zeta \sigma_{ir} d\zeta \quad (i \in \{u, v\}). \quad (26)$$

We shall also need the transverse shear *force* per unit width,

$$Q_i := \int_{-t/2}^{t/2} \sigma_{ir} d\zeta = t \bar{\sigma}_{ir} \quad (i \in \{u, v\}), \quad (27)$$

because it is what the boundary term of the moment integral produces. Multiplying Eq. (20) by ζ and integrating term by term, the two in-plane gradients give M_{uu} and M_{uv} directly, the two curvature terms give Ξ_{ur} and Ξ_{vr} directly, and the ζ -derivative integrates by parts to

$$\int_{-t/2}^{t/2} \zeta \frac{\partial \sigma_{ur}}{\partial \zeta} d\zeta = \left[\zeta \sigma_{ur} \right]_{-t/2}^{t/2} - \int_{-t/2}^{t/2} \sigma_{ur} d\zeta = \frac{t}{2} \left[\sigma_{ur} \left(\frac{t}{2} \right) + \sigma_{ur} \left(-\frac{t}{2} \right) \right] - Q_u, \quad (28)$$

the bracket picking up a plus sign because the lower limit carries $\zeta = -t/2$ against the outward orientation. Collecting,

$$\boxed{\frac{\partial M_{uu}}{\partial u} + \frac{\partial M_{uv}}{\partial v} + \kappa_{uu}\Xi_{ur} - \kappa_{uv}\Xi_{vr} = Q_u - \frac{t}{2} [\sigma_{ur}(\frac{t}{2}) + \sigma_{ur}(-\frac{t}{2})]}, \quad (29)$$

with the companion from Eq. (21),

$$\boxed{\frac{\partial M_{uv}}{\partial u} + \frac{\partial M_{vv}}{\partial v} - \kappa_{uv}\Xi_{ur} + \kappa_{vv}\Xi_{vr} = Q_v - \frac{t}{2} [\sigma_{vr}(\frac{t}{2}) + \sigma_{vr}(-\frac{t}{2})]}. \quad (30)$$

This is the shell analogue of the elementary beam relation between a moment gradient and a transverse shear force.

A trap to avoid here. It is tempting to apply assumption 5 to σ_{ur} and σ_{vr} at this point. Under a linear profile the surface mean equals the average, $\frac{1}{2} [\sigma_{ur}(\frac{t}{2}) + \sigma_{ur}(-\frac{t}{2})] = \bar{\sigma}_{ur}$, so the right-hand side of Eq. (29) collapses to $Q_u - t\bar{\sigma}_{ur} = 0$ and the moment balance appears to become homogeneous. That reduction is invalid, and the reason is elementary: a function linear in ζ that vanishes at both $\zeta = \pm t/2$ is identically zero, so a linear transverse-shear profile forces $Q_i = 0$ outright whenever the contact faces are shear-free. The true profile must be at least parabolic; Section 11.7 solves for it and finds exactly a parabola, $\sigma_{zr} \propto t^2/4 - \zeta^2$, which vanishes at both faces while carrying a nonzero resultant. Assumption 5 is therefore restricted to the in-plane stresses and σ_{rr} , and never applied to the transverse shears. **Downstream consequence:** the homogeneous pair that would follow — $\kappa_{uu}\Xi_{ur} = \kappa_{uv}\Xi_{vr}$ and its companion, whose determinant is $K = 0$ — is *not* a consequence of the moment balance, so no conclusion may be drawn from its singularity. Eq. (128) replaces it, and determines Q_z rather than constraining anything. The parity argument of Section 6.3 does not use the moment balance and is unaffected; only its closing sentence needs this amendment.

On the faces themselves the bracket is set by the interface, not by these equations: shear-free contact makes it vanish and leaves $\frac{\partial M_{uu}}{\partial u} + \frac{\partial M_{uv}}{\partial v} + \kappa_{uu}\Xi_{ur} - \kappa_{uv}\Xi_{vr} = Q_u$, which is the form Section 11.6 uses, while sliding friction makes it $-\mu t \bar{\sigma}_{rr} f_u$ by Eq. (49).

4 Constitutive relations and compatibility

4.1 Hooke's law

Because the ribbon is thin and its long edges are traction-free, the in-plane state is plane *stress*. For a linear isotropic material,

$$\bar{\epsilon}_{uu} = \frac{1}{E} (\bar{\sigma}_{uu} - \nu \bar{\sigma}_{vv}), \quad \bar{\epsilon}_{vv} = \frac{1}{E} (\bar{\sigma}_{vv} - \nu \bar{\sigma}_{uu}), \quad \bar{\epsilon}_{uv} = \frac{1+\nu}{E} \bar{\sigma}_{uv}, \quad (31)$$

inverted where convenient as

$$\bar{\sigma}_{uu} = \frac{E}{1-\nu^2} (\bar{\epsilon}_{uu} + \nu \bar{\epsilon}_{vv}), \quad \bar{\sigma}_{vv} = \frac{E}{1-\nu^2} (\bar{\epsilon}_{vv} + \nu \bar{\epsilon}_{uu}), \quad \bar{\sigma}_{uv} = \frac{E}{1+\nu} \bar{\epsilon}_{uv}. \quad (32)$$

4.2 Compatibility, with no curvature source

Equilibrium constrains forces to balance; it says nothing about whether the resulting strain field corresponds to an actual continuous displacement of an unbroken sheet. For a strain field on a surface of Gaussian curvature K , the linearized (von Kármán) compatibility relation carries $-K$ on its right-hand side. Here $K = 0$ exactly (Eq. 12), so the relation reduces to the flat Saint-Venant statement with no source term:

$$\boxed{\frac{\partial^2 \bar{\epsilon}_{uu}}{\partial v^2} + \frac{\partial^2 \bar{\epsilon}_{vv}}{\partial u^2} - 2 \frac{\partial^2 \bar{\epsilon}_{uv}}{\partial u \partial v} = 0.} \quad (33)$$

It is worth stating plainly what this rules out. One might hope that the tightening of the coil geometrically *forces* some membrane strain, which would supply the stress field with a source and so determine its magnitude. It does not: the tightening is isometric, so Eq. (33) is homogeneous, and the question of what sets the magnitude of the friction-driven stress is left open until Section 8.3.

Substituting Eq. (31) into Eq. (33) and multiplying through by E gives:

$$\boxed{\frac{\partial^2 \bar{\sigma}_{uu}}{\partial v^2} - \nu \frac{\partial^2 \bar{\sigma}_{vv}}{\partial v^2} + \frac{\partial^2 \bar{\sigma}_{vv}}{\partial u^2} - \nu \frac{\partial^2 \bar{\sigma}_{uu}}{\partial u^2} - 2(1 + \nu) \frac{\partial^2 \bar{\sigma}_{uv}}{\partial u \partial v} = 0.} \quad (34)$$

4.3 Equivalent displacement form

Eq. (34) is the awkward member of the governing set, and there is an exactly equivalent formulation that avoids it. Let $a(u, v)$ and $b(u, v)$ be the in-plane displacements along $\hat{u}$ and $\hat{v}$, so that

$$\bar{\epsilon}_{uu} = \frac{\partial a}{\partial u}, \quad \bar{\epsilon}_{vv} = \frac{\partial b}{\partial v}, \quad \bar{\epsilon}_{uv} = \frac{1}{2} \left(\frac{\partial a}{\partial v} + \frac{\partial b}{\partial u} \right). \quad (35)$$

Compatibility is then satisfied *identically*: it is nothing more than the statement that a displacement field exists, which Eq. (35) presumes. Writing the thickness-averaged in-plane equilibrium equations (23)–(24) with everything that is not an in-plane stress gradient moved to the right,

$$\frac{\partial \bar{\sigma}_{uu}}{\partial u} + \frac{\partial \bar{\sigma}_{uv}}{\partial v} = f_u^b, \quad \frac{\partial \bar{\sigma}_{uv}}{\partial u} + \frac{\partial \bar{\sigma}_{vv}}{\partial v} = f_v^b, \quad (36)$$

the pair (f_u^b, f_v^b) acts as an in-plane body force. Substituting Eqs. (32)–(35) turns Eq. (36) into the Navier equations:

$$\frac{E}{1 - \nu^2} \frac{\partial^2 a}{\partial u^2} + \frac{E}{2(1 + \nu)} \frac{\partial^2 a}{\partial v^2} + \left[\frac{E\nu}{1 - \nu^2} + \frac{E}{2(1 + \nu)} \right] \frac{\partial^2 b}{\partial u \partial v} = f_u^b, \quad (37)$$

$$\frac{E}{2(1 + \nu)} \frac{\partial^2 b}{\partial u^2} + \frac{E}{1 - \nu^2} \frac{\partial^2 b}{\partial v^2} + \left[\frac{E\nu}{1 - \nu^2} + \frac{E}{2(1 + \nu)} \right] \frac{\partial^2 a}{\partial u \partial v} = f_v^b. \quad (38)$$

These are two equations for two unknowns, second order, with the body force appearing undifferentiated. This is the form the numerical work of Section 9 uses, and Section 8.2 verifies that it is equivalent to Eq. (34).

5 Friction

5.1 Overlapping neighbours

Because the ribbon is wound around itself, an element at (u, v) can be in contact with another element one full turn away along the ribbon: at $(u^\pm, v^\pm)$, with equal z and θ differing by $\pm 2\pi$. Solving Eq. (8) for those two conditions,

$$\boxed{u^\pm = u \pm 2\pi r \cos \phi, \quad v^\pm = v \mp 2\pi r \sin \phi.} \quad (39)$$

Neglecting edge effects where $u^\pm$ falls outside $[0, L]$, an inner neighbour exists exactly when v^- lies within the ribbon's width, and likewise for the outer:

$$\text{inner neighbour present} \iff v \leq \frac{W}{2} - 2\pi r \sin \phi, \quad (40)$$

$$\text{outer neighbour present} \iff v \geq -\frac{W}{2} + 2\pi r \sin \phi. \quad (41)$$

5.2 Slip kinematics and the friction direction

Coulomb friction opposes relative sliding, so we need the relative velocity of two elements in contact as x advances. Differentiating Eq. (8) at fixed (u, v) ,

$$\frac{\partial r}{\partial x} = \frac{dr}{dx}, \quad \frac{\partial \theta}{\partial x} = -\frac{\theta}{r} \frac{dr}{dx} - \frac{u \sin \phi + v \cos \phi}{r} \frac{d\phi}{dx}, \quad \frac{\partial z}{\partial x} = (u \cos \phi - v \sin \phi) \frac{d\phi}{dx}. \quad (42)$$

Subtracting the same quantities at the inner contact partner (Eq. 39), converting the angular part to an arc-length rate by multiplying by r , and using $\theta(u^-, v^-) - \theta(u, v) = -2\pi$ exactly, the position-dependent pieces cancel and

$$\Delta v_\theta = -2\pi \frac{dr}{dx}, \quad \Delta v_z = 2\pi r \frac{d\phi}{dx}. \quad (43)$$

Both are independent of (u, v) : every contact in the coil slides at the same rate. Projecting onto the surface frame using the rotation by ϕ ,

$$\vec{v}_{\text{slip}} = 2\pi \begin{pmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{pmatrix} \begin{pmatrix} -\frac{dr}{dx} \\ r \frac{d\phi}{dx} \end{pmatrix}, \quad (44)$$

and defining (f_u, f_v) as the unit vector along the slip direction,

$$f_u = \frac{-\frac{dr}{dx} \cos \phi + r \frac{d\phi}{dx} \sin \phi}{\sqrt{\left(\frac{dr}{dx}\right)^2 + r^2 \left(\frac{d\phi}{dx}\right)^2}}, \quad f_v = \frac{\frac{dr}{dx} \sin \phi + r \frac{d\phi}{dx} \cos \phi}{\sqrt{\left(\frac{dr}{dx}\right)^2 + r^2 \left(\frac{d\phi}{dx}\right)^2}}. \quad (45)$$

In the inextensible limit this collapses completely. There $r = r_{\text{nat}} \cos \phi$, so $\frac{dr}{dx} = -r_{\text{nat}} \sin \phi \frac{d\phi}{dx}$, and on substituting into Eq. (45) both $\frac{d\phi}{dx}$ and r_{nat} cancel between numerator and denominator, leaving

$$\boxed{(f_u, f_v) = (\sin 2\phi, \cos 2\phi).} \quad (46)$$

We use this form from here on, and note the one place it degenerates: at $x = 0$ exactly, $\frac{dr}{dx} = 0$ and $\phi = 0$, so Eq. (46) gives $(f_u, f_v) = (0, 1)$ — the only sliding at the very start of the pull is axial, with no circumferential component at all. Since $\Delta v_z = 2\pi r \frac{d\phi}{dx} \neq 0$ there, sliding is nonetheless active for every $x > 0$, which is what justifies using a kinetic friction law throughout (assumption 6, Section 1.2).

5.3 Friction magnitude

The friction exerted on an element by the layer inside it resists that element's forward motion, and its components are the shear stresses on the inner $(-\hat{r})$ face. Coulomb's law makes the magnitude μ times the normal pressure $-\sigma_{rr}^-$, so

$$\sigma_{ur}\left(-\frac{t}{2}\right) = -\mu \sigma_{rr}^- f_u, \quad \sigma_{vr}\left(-\frac{t}{2}\right) = -\mu \sigma_{rr}^- f_v. \quad (47)$$

The friction from the outer layer is the exact opposite in physical direction; because the outward normal of that face is $+\hat{r}$ rather than $-\hat{r}$, the sign convention for shear stresses makes the stress components take the same form,

$$\sigma_{ur}\left(\frac{t}{2}\right) = -\mu \sigma_{rr}^+ f_u, \quad \sigma_{vr}\left(\frac{t}{2}\right) = -\mu \sigma_{rr}^+ f_v. \quad (48)$$

Each pair holds only where the corresponding neighbour is present (Eqs. 40–41); where it is absent, both the shear and the pressure that would drive it vanish. Taking the mean and the difference of Eqs. (47)–(48),

$$\boxed{\bar{\sigma}_{ur} = -\mu \bar{\sigma}_{rr} f_u, \quad \bar{\sigma}_{vr} = -\mu \bar{\sigma}_{rr} f_v,} \quad (49)$$

$$\Delta \sigma_{ur} = -\mu \Delta \sigma_{rr} f_u, \quad \Delta \sigma_{vr} = -\mu \Delta \sigma_{rr} f_v. \quad (50)$$

5.4 Layer-contact recursion

By Newton's third law, two surfaces in contact carry equal and opposite tractions; since their normals are opposite, the stress components are equal. It suffices to equate the normal pressures since the shears are tied to them by Eq. (49). This gives the recursive relationship

$$\text{outer surface: } \sigma_{rr}^+(u, v) = \begin{cases} \sigma_{rr}^-(u^+, v^+) & \text{outer neighbour present} \\ 0 & \text{otherwise} \end{cases} \quad (51)$$

$$\text{inner surface: } \sigma_{rr}^-(u, v) = \begin{cases} \sigma_{rr}^+(u^-, v^-) & \text{inner neighbour present} \\ 0 & \text{otherwise} \end{cases} \quad (52)$$

Propagated far enough in u , Eq. (51) reaches the outermost wound layer near $u = L$, past which no material is wound at all, so $\sigma_{rr}^+ = 0$ there. That is the base case that starts the recursion, and it is a geometric fact rather than an imposed boundary condition: nothing is wound outside the last turn, independent of how the tip happens to be gripped.

Eqs. (51)–(52) are exact, and the numerical work of Section 9 uses them in that form. However, for analytical work, it is convenient to Taylor-expand in the shift of Eq. (39), which in the case where both neighbours are present gives the differential relation

$$\boxed{\Delta\sigma_{rr} = 2\pi r \cos\phi \frac{\partial\bar{\sigma}_{rr}}{\partial u} - 2\pi r \sin\phi \frac{\partial\bar{\sigma}_{rr}}{\partial v}.} \quad (53)$$

This is a first-order transport equation: it advects $\bar{\sigma}_{rr}$ along the direction $(\cos\phi, -\sin\phi)$ in the (u, v) plane, driven by $\Delta\sigma_{rr}$. Its characteristic direction is the ribbon's own winding direction, as it must be, since the recursion steps from one turn to the next.

6 Bending resistance

We now account for the material's resistance to bending. This component is an independent physical effect, and can be fully described without introducing any undetermined constants.

6.1 Natural curvature and the mismatch

A sheet that has been wound and left alone acquires a preferred curl. We take that natural state to be a cylinder of radius $r_{\text{nat}} = L/\theta_L$ at zero pitch, i.e., the curvature triple $\kappa_{\text{nat}} = [1/r_{\text{nat}}, 0, 0]$ in the ordering $[\kappa_{uu}, \kappa_{uv}, \kappa_{vv}]$. The mismatch between actual and natural curvature is then

$$\boxed{\Delta\kappa_{uu} = \frac{\cos^2\phi}{r} - \frac{1}{r_{\text{nat}}}, \quad \Delta\kappa_{uv} = \frac{\sin\phi \cos\phi}{r}, \quad \Delta\kappa_{vv} = \frac{\sin^2\phi}{r},} \quad (54)$$

and it satisfies two identities:

$$\Delta\kappa_{uu} + \Delta\kappa_{vv} = \frac{1}{r} - \frac{1}{r_{\text{nat}}}, \quad \Delta\kappa_{uu}\Delta\kappa_{vv} - \Delta\kappa_{uv}^2 = -\frac{\sin^2\phi}{r r_{\text{nat}}}. \quad (55)$$

The first is immediate from $\cos^2\phi + \sin^2\phi = 1$. In the second, the cross terms cancel, leaving $-\sin^2\phi/(r r_{\text{nat}})$, which is nonzero for any $\phi \neq 0$. Note that this is the determinant of the *mismatch*, not of the curvature itself, so there is no contradiction with $K = 0$ (Eq. 12).

At $x = 0$, we have $\phi = 0$ and $\rho = 1$, so all three components of Eq. (54) vanish and the sheet is unstressed as intended. Note also that each component depends on x alone, through $\phi(x)$ and $r(x)$, with no dependence on (u, v) .

6.2 Decoupling bending moments

The Kirchhoff–Love closure relates the moments of Eq. (26) to the curvature mismatch,

$$\begin{aligned} M_{uu} &= D [(1 - \nu) \Delta \kappa_{uu} + \nu (\Delta \kappa_{uu} + \Delta \kappa_{vv})], \\ M_{vv} &= D [(1 - \nu) \Delta \kappa_{vv} + \nu (\Delta \kappa_{uu} + \Delta \kappa_{vv})], \\ M_{uv} &= D (1 - \nu) \Delta \kappa_{uv}, \end{aligned} \quad D := \frac{E t^3}{12 (1 - \nu^2)}. \quad (56)$$

Since Eq. (54) depends only on x , so do all three moments. Section 11.6 derives Eq. (56) from the three-dimensional constitutive law rather than positing it, and finds one additional term driven by the pressure difference across the thickness; that term is the only (u, v) dependence the moments carry, and it is what makes the moment gradients in Eqs. (29)–(30) nonzero.

6.3 Why bending does not move the contact pressure

It is natural to expect the curvature mismatch to enter through the same channel as everything else — as an addition to the surface pressures $\sigma_{rr}^\pm$, which is what drives friction. It does not, and the reason is a parity argument worth spelling out.

Let the in-plane stresses carry both a membrane (mean across the thickness) and a bending part (deviation from the mean):

$$\sigma_{ij}(\zeta) = \bar{\sigma}_{ij} + \zeta \sigma'_{ij}, \quad (57)$$

Now integrate the exact pointwise radial equilibrium equation (22) in ζ , starting from the true surface value σ_{rr}^- . Write C_0 for the membrane part of the right-hand side:

$$C_0 := -\frac{\partial \bar{\sigma}_{ur}}{\partial u} - \frac{\partial \bar{\sigma}_{vr}}{\partial v} + \kappa_{uu} \bar{\sigma}_{uu} - 2\kappa_{uv} \bar{\sigma}_{uv} + \kappa_{vv} \bar{\sigma}_{vv}, \quad (58)$$

Integrating again,

$$\sigma_{rr}(\zeta) = \sigma_{rr}^- + C_0 \left(\zeta + \frac{t}{2} \right) + \frac{6}{t^3} [\kappa_{uu} M_{uu} - 2\kappa_{uv} M_{uv} + \kappa_{vv} M_{vv}] \left(\zeta^2 - \frac{t^2}{4} \right). \quad (59)$$

The bending correction is proportional to $\zeta^2 - t^2/4$, which vanishes at *both* surfaces: at $\zeta = -t/2$ by construction of the integration constant, and at $\zeta = +t/2$ because $(t/2)^2 - t^2/4 = 0$ as well. It can bow the pressure profile in the interior of the thickness but can never shift either surface value. Hence

$$\Delta \sigma_{rr} = C_0 t = t [\kappa_{uu} \bar{\sigma}_{uu} - 2\kappa_{uv} \bar{\sigma}_{uv} + \kappa_{vv} \bar{\sigma}_{vv}] + t\mu \left[\sin 2\phi \frac{\partial \bar{\sigma}_{rr}}{\partial u} + \cos 2\phi \frac{\partial \bar{\sigma}_{rr}}{\partial v} \right], \quad (60)$$

where the second bracket comes from substituting Eq. (49) into the shear-gradient terms of C_0 , ϕ being uniform so that the derivative passes through $\sin 2\phi$ and $\cos 2\phi$. We retain this term rather than dropping it, although it is smaller than the first bracket by a factor of order t/r ; Section 8.6 identifies where it is safe to neglect.

So $\Delta \sigma_{rr}$ is unaffected by bending, and since neither surface value moves, $\bar{\sigma}_{rr}$ is unaffected too.

The conclusion is that in this geometry a spatially uniform curvature mismatch is invisible to the contact pressure field: the friction law (49), the recursion (51)–(52) and the transport equation (53) all stand with no bending correction. Bending does contribute to $F(x)$, but through stored energy (Section 6.4) rather than through friction.

6.4 Bending energy

The Kirchhoff–Love energy density is $u_{\text{bend}} = \frac{1}{2} M_{ij} \Delta \kappa_{ij}$, which with Eq. (56) is the standard isotropic form

$$u_{\text{bend}} = \frac{D}{2} \left[(\Delta \kappa_{uu} + \Delta \kappa_{vv})^2 - 2(1 - \nu) (\Delta \kappa_{uu} \Delta \kappa_{vv} - \Delta \kappa_{uv}^2) \right]. \quad (61)$$

Substituting the two identities of Eq. (55) collapses this to a closed form in r and ϕ alone,

$$u_{\text{bend}}(r, \phi) = \frac{D}{2} \left[\left(\frac{1}{r} - \frac{1}{r_{\text{nat}}} \right)^2 + \frac{2(1 - \nu) \sin^2 \phi}{r r_{\text{nat}}} \right], \quad (62)$$

with total stored energy, the density being uniform over the sheet,

$$U_{\text{bend}}(x) = L W u_{\text{bend}}(r(x), \phi(x)). \quad (63)$$

The two terms have distinct characters. The first is a *hoop* mismatch: it penalizes the coil radius differing from the natural radius, and vanishes when $r = r_{\text{nat}}$. The second is a *twist* mismatch, traceable to $\Delta \kappa_{uv}$ through the determinant identity; it penalizes pitch and vanishes only at $\phi = 0$. At small ϕ the twist term dominates: expanding Eq. (54), $\Delta \kappa_{uv} = O(\phi)$ whereas $\Delta \kappa_{uu}$ and $\Delta \kappa_{vv}$ are both $O(\phi^2)$. That observation reappears in Section 7.4 as the reason the initial branch of $F(x)$ is soft and linear.

7 Coupling I: the coil radius

We now combine the above governing relationships, beginning with the question: given x , what r minimizes the elastic energy in the material? In other words, what r balances the bending penalty of Eq. (62) against the stretching penalty of Eq. (64)?

7.1 The two-mode energy

The membrane energy is based on the mean strain ϵ . The long edges are traction-free, the state is uniaxial *stress* rather than uniaxial strain, so the density is $\frac{1}{2} E \epsilon^2$ per unit volume with no $1/(1 - \nu^2)$ stiffening:

$$U_{\text{mem}} = \frac{1}{2} E t L W \epsilon^2. \quad (64)$$

Adding the bending energy component of Eq. (63), substituting the nondimensional variables of Eqs. (5) and factoring out the common scale, the total collapses onto a single dimensionless function

of one variable:

$$U(\rho; \chi) = \frac{L W D}{2 r_{\text{nat}}^2} \tilde{U}(\rho; \chi), \quad \tilde{U} = \underbrace{\left(\frac{1}{\rho} - 1\right)^2}_{\text{bending}} + \underbrace{\frac{2(1-\nu)\chi^2}{\rho\lambda^2}}_{\text{stretching}} + \underbrace{\Gamma(\lambda-1)^2}_{\text{stretching}} \quad (65)$$

with the whole competition carried by one dimensionless group:

$$\Gamma := \frac{E t r_{\text{nat}}^2}{D} = \frac{12(1-\nu^2) r_{\text{nat}}^2}{t^2}. \quad (66)$$

Observe that the Young's modulus cancels out. This means the balance point between bending and stretching is therefore purely geometric, fixed by t/r_{nat} and ν alone.

Two scalars stand in for the whole stress state here, which is a strong reduction, and it is presented as one. Section 11 derives Eq. (65) instead, starting from the general elastic energy density $\frac{1}{2}\sigma_{ij}\epsilon_{ij}$ with all twelve of its unknown fields and eliminating them one at a time; it recovers this equation exactly, and identifies which assumptions each elimination spends. Nothing below depends on that derivation, but it is what licenses the reduction.

7.2 Exact stationarity

Throughout this section “stationary” means stationary with respect to ρ *at fixed* χ . Eq. (67) below holds separately at every extension, and its root $\rho(\chi)$ therefore varies with χ ; nothing is being held fixed in x . This distinction is what makes Section 7.4 work.

Differentiating Eq. (65) at fixed χ , using $\frac{\partial \lambda}{\partial \rho} = \rho/\lambda$, the condition $\frac{\partial \tilde{U}}{\partial \rho} = 0$ simplifies to the form

$$\underbrace{\frac{1-\rho}{\rho^3}}_{\text{hoop}} + \underbrace{\frac{(1-\nu)\chi^2(3\rho^2+\chi^2)}{\rho^2\lambda^4}}_{\text{twist}} = \underbrace{\frac{\Gamma\rho(\lambda-1)}{\lambda}}_{\text{membrane}}. \quad (67)$$

Both left-hand terms are strictly positive for $0 < \rho < 1$: *both* bending contributions push the radius up, since a larger r means both less hoop mismatch and — because $\sin \phi = \chi/\lambda$ falls as ρ grows — less twist. The membrane term pushes it down. The left side diverges as $\rho \rightarrow 0$ and vanishes as $\rho \rightarrow 1$ with $\chi \rightarrow 0$, so there is a unique interior root (Figure 3b).

That root has no useful closed form. Clearing the denominators of Eq. (67) and squaring away the single radical $\lambda = \sqrt{\rho^2 + \chi^2}$ leaves a polynomial of degree sixteen in ρ , which is neither solvable in radicals nor informative if it were. What makes this a non-issue is that an accurate closed form is available by a different route, developed below: because Γ is large, the root can be obtained by perturbation rather than by inversion.

One consequence deserves emphasis. The right-hand side vanishes only at $\lambda = 1$, where the left-hand side does not. So $\lambda > 1$ strictly for every $\chi > 0$. In other words, **the ribbon is always slightly stretched**, and “inextensible” is a limit that is never exactly attained. This residual strain, small as it is, is the only non-homogeneous source of stress to tie the friction field to a

definite magnitude (Section 8.3).

Since Γ is large, Eq. (67) can be solved by perturbation in $1/\Gamma$ rather than by inversion. Note that this expansion is in $1/\Gamma$ and *not* in ρ , so it makes no assumption about how far the coil has tightened. The argument takes three steps.

Step 1: the $\Gamma \rightarrow \infty$ limit. Rearrange Eq. (67) so that the only factor carrying Γ stands alone:

$$\frac{\lambda - 1}{\lambda} = \frac{1}{\Gamma \rho} \left[\frac{1 - \rho}{\rho^3} + \frac{(1 - \nu) \chi^2 (3\rho^2 + \chi^2)}{\rho^2 \lambda^4} \right]. \quad (68)$$

For any ρ in $(0, 1)$ the bracket is finite, so the entire right-hand side is of order Γ^{-1} , and therefore

$$\lambda - 1 = O(\Gamma^{-1}) \quad \implies \quad \lambda \rightarrow 1 \quad \text{as} \quad \Gamma \rightarrow \infty. \quad (69)$$

An infinitely stiff membrane is unstretched, as it must be. Now $\lambda^2 = \rho^2 + \chi^2$ by Eq. (5), so imposing $\lambda = 1$ forces $\rho^2 + \chi^2 = 1$. Call the resulting radius

$$\boxed{\rho_0 := \sqrt{1 - \chi^2}} \quad (70)$$

so that ρ_0 is, by construction, the root of Eq. (67) in the limit of infinite membrane stiffness. It is not a new assumption. By Eq. (5), $\lambda = 1$ is the same statement as $\epsilon = 0$, and in that limit $\cos \phi = \rho/\lambda \rightarrow \rho_0$ while $\sin \phi = \chi/\lambda \rightarrow \chi$, so $\rho_0 = \cos \phi$ with $\chi = \sin \phi$ — precisely the inextensible relations of Eq. (4). The $\Gamma \rightarrow \infty$ limit of the energy balance and the inextensible kinematics of Section 2.1 are therefore the same statement, which is the sense in which those relations were a limit rather than an identity.

Step 2: expand both sides about that limit. Let Γ be large but finite, and write $\lambda = 1 + \epsilon$ with $\epsilon = O(\Gamma^{-1})$ from Eq. (69). On the right-hand side of Eq. (67),

$$\frac{\Gamma \rho (\lambda - 1)}{\lambda} = \frac{\Gamma \rho \epsilon}{1 + \epsilon} = \Gamma \rho \epsilon [1 - \epsilon + \dots] = \Gamma \rho_0 \epsilon + O(\Gamma^{-1}), \quad (71)$$

where the two corrections absorbed into $O(\Gamma^{-1})$ are $\Gamma \rho \epsilon^2$, which is $O(\Gamma^{-1})$ because $\epsilon^2 = O(\Gamma^{-2})$, and the replacement of ρ by ρ_0 , which by Eq. (74) below costs $\Gamma \epsilon \cdot O(\Gamma^{-1})$, also $O(\Gamma^{-1})$.

On the left-hand side, the two bending terms form a smooth function of ρ and λ near $(\rho_0, 1)$, so they may simply be evaluated there, again incurring an error $O(\Gamma^{-1})$. Setting $\lambda = 1$ makes $\lambda^4 = 1$, and $\chi^2 = 1 - \rho_0^2$ eliminates χ ; the latter also collapses the twist numerator, since $3\rho_0^2 + \chi^2 = 3\rho_0^2 + 1 - \rho_0^2 = 1 + 2\rho_0^2$. The left-hand side is therefore

$$\underbrace{\frac{1 - \rho_0}{\rho_0^3}}_{\text{hoop}} + \underbrace{\frac{(1 - \nu) (1 - \rho_0^2) (1 + 2\rho_0^2)}{\rho_0^2}}_{\text{twist}}. \quad (72)$$

Step 3: solve for the strain. Equating Eqs. (71) and (72) and dividing through by $\Gamma\rho_0$,

$$\epsilon(\chi) = \frac{(1 - \rho_0) + (1 - \nu) \rho_0 (1 - \rho_0^2) (1 + 2\rho_0^2)}{\Gamma \rho_0^4} + O(\Gamma^{-2}), \quad \rho_0 = \sqrt{1 - \chi^2}, \quad (73)$$

where the common factor ρ_0^{-4} has been taken outside, which is what turns the ρ_0^{-3} and ρ_0^{-2} of Eq. (72) into the ρ_0 and ρ_0^0 weights seen inside the bracket. Here $O(\Gamma^{-2})$ is the usual order symbol, denoting the neglected terms: they shrink at least as fast as Γ^{-2} as Γ grows, and so are smaller than the term retained by a factor of order $\Gamma^{-1} \sim 10^{-6}$ for the geometry of Section 9.2. Finally the radius follows from $\lambda = 1 + \epsilon$ together with $\lambda^2 = \rho^2 + \chi^2$,

$$\rho^2 = (1 + \epsilon)^2 - \chi^2 = \rho_0^2 + 2\epsilon + \epsilon^2 \quad \implies \quad \rho(\chi) = \sqrt{\rho_0^2 + 2\epsilon(\chi)}, \quad (74)$$

the dropped ϵ^2 being $O(\Gamma^{-2})$ once more. Eqs. (73)–(74) are the closed form promised above: ρ_0 from the geometry, ϵ from Eq. (73), and ρ from Eq. (74), with no root-finding anywhere.

The bracket in Eq. (73) is slowly varying and of order unity across the whole relevant range: it equals 1 at $\rho_0 = 0$, 0.93 at $\rho_0 = 0.3$ and 0.89 at $\rho_0 = 0.5$ (taking $\nu = 0.3$), falling to zero only as $\rho_0 \rightarrow 1$, where the coil is unpulled and there is no bending drive to balance. That near-constancy is a useful robustness statement, because a sheet whose rest state is *flat* rather than curled — i.e. $1/r_{\text{nat}} \rightarrow 0$ in Eq. (54) — has $u_{\text{bend}} = D/2r^2$ exactly, with no twist term and no determinant contribution, and reproduces Eq. (73) with the bracket replaced by 1. The two bracketing choices of natural curvature therefore differ by at most about eleven percent.

Figure 2 checks Eq. (73) against the exact root of Eq. (67) obtained numerically. The relative error in ϵ is 1.6×10^{-4} at $x/L = 0.8$ and 1.2×10^{-3} at $x/L = 0.9$, first exceeding 1% at $x/L \approx 0.95$ and then climbing steeply. So the closed form is accurate to a fraction of a percent over essentially the whole pull, and degrades only as the extension approaches the crossover x_* identified in Section 7.3 — which is exactly where the expansion should fail, since $\delta \ll \rho_0^2$ is the assumption that defines x_* . That failure is therefore not a nuisance but the same statement as Eq. (78). For practical purposes Eq. (73) *is* the solution up to the crossover, and the numerical root only confirms it.

Panel (a) also displays the feature of $\epsilon(\chi)$ most worth internalizing. The strain is of order 10^{-7} at small extension and climbs through four orders of magnitude, reaching a few times 10^{-3} only as the coil stops tightening. The ribbon is *barely* stretched over almost the whole pull — but it is never exactly unstretched, and Section 8.4 shows that this is the entire origin of the friction force.

7.3 The sliding-to-stretching crossover

Section 7.2 left two things unsettled: the coil tightens as it is pulled, but the ribbon also stretches slightly at every extension, so at what point does one give way to the other? And what stops the tightening?

There is no floor. It is tempting to look for a smallest radius the coil can reach. There is none. Solving Eq. (67) numerically over the whole range shows ρ decreasing *monotonically* in χ , with no turning point: $\frac{d\rho}{d\chi} < 0$ everywhere. Pushed far past $x = L$ — which the ribbon permits, since

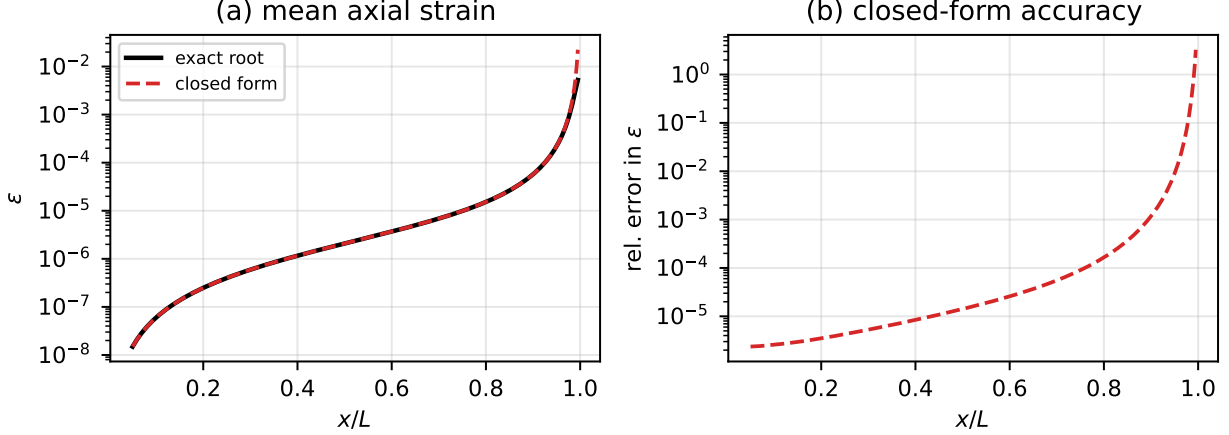


Figure 2: (a) The mean axial strain, on a logarithmic scale: the closed form Eq. (73) against the exact root of Eq. (67). The two are indistinguishable over almost the whole range. The strain is tiny but strictly positive everywhere, and rises steeply only in the last few percent of extension. (b) Relative error of the closed form: below 10^{-3} out to $x/L \approx 0.89$ and below 1% out to $x/L \approx 0.95$, rising sharply thereafter as the extension approaches the crossover of Eq. (78), where the expansion's own validity condition fails.

$x \leq \Lambda = L(1 + \epsilon)$ — the balance in Eq. (67) between the hoop term ρ^{-3} and the membrane term settles on

$$\rho \approx \left[\frac{\chi}{\Gamma(\chi - 1)} \right]^{1/4} \xrightarrow{\chi \gg 1} \rho \rightarrow \Gamma^{-1/4}, \quad (75)$$

so the radius approaches a small value asymptotically but never attains it and never turns around. What exists is therefore not a floor but a change of regime, and it needs defining in a way that does not smuggle in an arbitrary threshold.

A threshold-free definition. Differentiate $\lambda^2 = \rho^2 + \chi^2$ (Eq. 5) with respect to χ :

$$\boxed{\underbrace{\lambda \frac{d\lambda}{d\chi}}_{\text{into stretching}} - \underbrace{\rho \frac{d\rho}{d\chi}}_{\text{into tightening}} = \chi} \quad (76)$$

This is an exact bookkeeping identity, and it says something useful: an increment of pull is spent partly on lengthening the ribbon (λ grows) and partly on tightening the coil (ρ shrinks, so the second term is positive), and the two shares always sum to χ . Both shares are therefore well defined at every extension, with no expansion and no threshold. Define the crossover $*$ as the extension at which they are *equal*, each taking half of the increment:

$$\lambda \left. \frac{d\lambda}{d\chi} \right|_* = -\rho \left. \frac{d\rho}{d\chi} \right|_* = \frac{\chi_*}{2}. \quad (77)$$

Below x_* the pull is absorbed almost entirely by tightening; above it, almost entirely by stretching. This is what “sliding gives way to stretching” means quantitatively, and it replaces the vaguer statement that the expansion of Section 7.2 stops being valid.

Evaluating Eq. (77) on the solution of Eq. (67) gives, to within a fraction of a percent across five decades of Γ ,

$$\boxed{\rho_* \approx \left(\frac{4}{\Gamma}\right)^{1/6}, \quad 1 - \frac{x_*}{L} \approx \frac{\rho_*^2}{4} = 2^{-4/3} \Gamma^{-1/3}, \quad \epsilon_* \approx \frac{\rho_*^2}{2}.} \quad (78)$$

In dimensional form the crossover radius is

$$r_* = r_{\text{nat}} \left(\frac{4}{\Gamma}\right)^{1/6} = \frac{(r_{\text{nat}}^2 t)^{1/3}}{(3(1-\nu^2))^{1/6}} \sim (r_{\text{nat}}^2 t)^{1/3}, \quad (79)$$

the geometric-mean scaling characteristic of bending–stretching boundary layers in shell theory. Both the crossover displacement and the crossover strain follow a $2/3$ power law in t/r_{nat} , and since Γ is independent of E (Section 7.1), neither depends on the sheet’s stiffness. Note that the $1/6$ and $2/3$ exponents are unchanged from what a cruder, threshold-based definition would give; only the prefactors move, which is a sign the scaling is a property of the balance rather than of the definition.

Packing is the same mechanism, not a second one. Eq. (13) gave a geometric limit ρ_{jam} at which the innermost turn closes on the axis, and it is natural to want to combine it with ρ_* . Writing $\rho_{\text{stop}} = \max(\rho_*, \rho_{\text{jam}})$ would be wrong on two counts: it treats packing as a hard wall, and it treats the two as alternatives when they should act together. Both objections have the same resolution, which is that *these are not two mechanisms*. Each is the bending energy diverging as some radius goes to zero — for ρ_* the mean radius, for ρ_{jam} the innermost one — and both can be captured by writing the bending energy over the true radius distribution instead of at the mean.

With turns stacked one thickness apart, the local radius at fractional position $\varsigma \in [-\frac{1}{2}, \frac{1}{2}]$ along the ribbon is

$$\rho_{\text{loc}}(\varsigma) = \rho + 2\rho_{\text{jam}} \varsigma, \quad (80)$$

running from $\rho - \rho_{\text{jam}}$ at the innermost turn to $\rho + \rho_{\text{jam}}$ at the outermost, which reproduces Eq. (13) exactly: the inner radius vanishes when $\rho = \rho_{\text{jam}}$. Averaging the two radius-dependent factors of Eq. (62) over ς gives closed forms,

$$\left\langle \frac{1}{\rho_{\text{loc}}^2} \right\rangle = \frac{1}{\rho^2 - \rho_{\text{jam}}^2}, \quad \left\langle \frac{1}{\rho_{\text{loc}}} \right\rangle = \frac{1}{2\rho_{\text{jam}}} \ln \left(\frac{\rho + \rho_{\text{jam}}}{\rho - \rho_{\text{jam}}} \right) =: \mathcal{L}, \quad (81)$$

and substituting them into Eq. (65) in place of ρ^{-2} and ρ^{-1} replaces that energy with

$$\boxed{\tilde{U} = \frac{1}{\rho^2 - \rho_{\text{jam}}^2} - 2\mathcal{L} + 1 + \frac{2(1-\nu)\chi^2}{\lambda^2} \mathcal{L} + \Gamma(\lambda - 1)^2.} \quad (82)$$

This is one smooth energy with no max anywhere in it. As $\rho_{\text{jam}} \rightarrow 0$, $\mathcal{L} \rightarrow 1/\rho$ and Eq. (82) reduces term by term to Eq. (65). As $\rho \rightarrow \rho_{\text{jam}}$ both $(\rho^2 - \rho_{\text{jam}}^2)^{-1}$ and $\mathcal{L}$ diverge, so the packing limit appears as an *asymptote* of the same bending energy rather than as an imposed constraint — the coil approaches it and never reaches it, exactly as it approaches Eq. (75) in the thin-stack limit.

The two contributions genuinely add. Minimizing Eq. (82) numerically at the parameters of Section 9.2, the radius attained near $x \approx L$ exceeds *both* ρ_* and ρ_{jam} through the region where they are comparable, by up to about 15% — a blend, not a switch. Which one sets the scale is a property of the winding alone. Since $\rho_{\text{jam}} = \frac{1}{2} n t / r_{\text{nat}}$ grows linearly in the number of turns while $\rho_* \propto (t/r_{\text{nat}})^{1/3}$ does not, their ratio is

$$\frac{\rho_{\text{jam}}}{\rho_*} = \frac{n}{2} \left(\frac{t}{r_{\text{nat}}} \right)^{2/3} (3(1 - \nu^2))^{1/6}, \quad (83)$$

so packing dominates for many turns or a thick ribbon, and the elastic mechanism dominates for few turns or a thin one, with the boundary $n \propto (t/r_{\text{nat}})^{-2/3}$. Figure 3(c) maps this out. The coil of Section 9.2 sits at $\rho_{\text{jam}}/\rho_* = 0.28$, well inside the elastically dominated region, which is why ρ_{jam} appears to do so little there; a coil wound with ten times as many turns would sit on the other side of the line. Because that ratio is 0.28, the numerical work of Section 9 uses the $\rho_{\text{jam}} \rightarrow 0$ limit Eq. (65) rather than Eq. (82); the correction factor $\rho^2 / (\rho^2 - \rho_{\text{jam}}^2)$ stays below 1.05 over the solved range.

What the crossover means for the experiment. Past x_* the pull goes mostly into strain, so the strain climbs steeply while the coil radius nearly stalls. This is the model’s account of lock-up, and it is a crossover rather than an event: nothing is pinned, nothing is discontinuous, and the coil is still tightening slowly on the far side of it. Whether a given sheet survives to display the change is a separate question — ϵ_* from Eq. (78) must lie below the material’s failure strain — and for paper the two are comparable, which is why paper tears rather than stretching visibly.

7.4 The conjugate force

With $\rho(x)$ determined, the force follows without further work, and with a simplification worth pointing out. Along the minimizing path,

$$F(x) = \frac{d}{dx} U(\rho(x); x) = \left. \frac{\partial U}{\partial x} \right|_{\rho} + \underbrace{\frac{\partial U}{\partial \rho} \frac{d\rho}{dx}}_{=0} = \left. \frac{\partial U}{\partial x} \right|_{\rho}, \quad (84)$$

the second term vanishing because $\frac{\partial U}{\partial \rho} = 0$ is exactly the stationarity condition (67). This is the envelope theorem, and the word “stationary” is doing specific work here: U is stationary against variations *of ρ at fixed x* , which is what kills the second term, while the root $\rho(x)$ itself certainly does vary with x — indeed $\frac{d\rho}{dx} \neq 0$, as Figure 3(a) shows. The content of Eq. (84) is that we never need its value. So $\frac{d\rho}{dx}$ is never needed: the work $F dx$ is the *partial* derivative at frozen radius, evaluated at the minimizer. One consequence is not obvious in advance — the hoop term

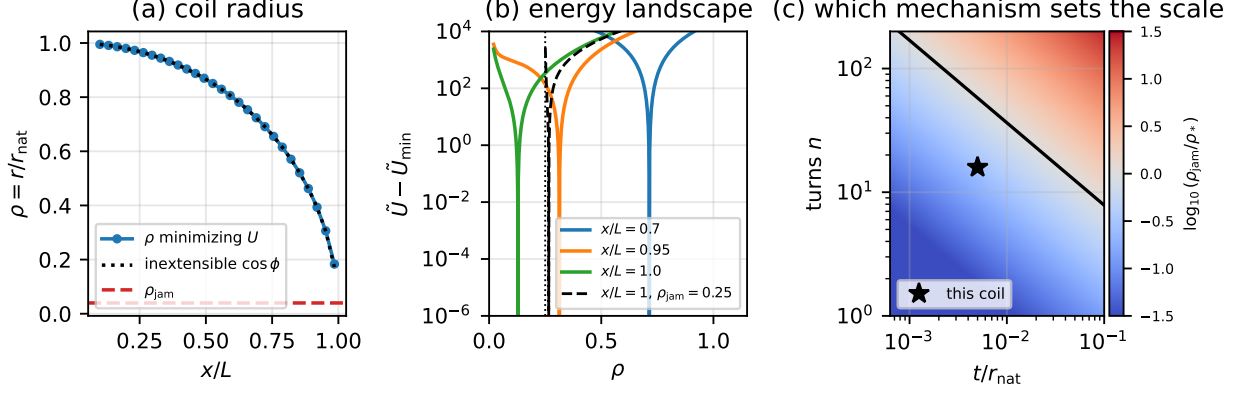


Figure 3: (a) The energy-minimizing radius from Eq. (67), against the inextensible $\cos \phi$ and the packing floor Eq. (13). The minimizer tracks the inextensible value closely, consistent with $\Gamma \gg 1$, and decreases monotonically throughout — there is no turning point. (b) The energy landscape of Eq. (82) at three extensions with $\rho_{\text{jam}} = 0$, plus one case with $\rho_{\text{jam}} = 0.25$ (dashed) showing the packing divergence appear as an asymptote of the same energy. The well is single and deep, so the minimization is unambiguous. (c) Which mechanism sets the radius scale, from Eq. (83): the black contour is $\rho_{\text{jam}} = \rho_*$, and the star marks the coil of Section 9.2.

$(1/\rho - 1)^2$ of Eq. (65) carries no explicit x , so it contributes *nothing* directly to the force, despite being the term that diverges as the coil tightens. It acts only indirectly, by setting where $\rho(x)$ sits. Differentiating the remaining two terms,

$$F(x) = \underbrace{\frac{2(1-\nu)WD}{r_{\text{nat}}^2} \frac{\rho\chi}{\lambda^4}}_{\text{twist (bending)}} + \underbrace{EtW\epsilon \frac{\chi}{\lambda}}_{\text{membrane tension}}, \quad (85)$$

the second term following from $\Gamma WD/r_{\text{nat}}^2 = EtW$ by Eq. (66). It is exactly the ribbon's axial tension $EtW\epsilon$ projected through $\chi/\lambda = \sin \phi$, as it must be. Both terms are of the same order in general: ϵ is small, of order $1/\Gamma$, but it is multiplied by the large Et .

Eq. (85) can be made independent of Γ altogether. Substituting $\Gamma(\lambda - 1)/\lambda$ from the stationarity condition (67) — an exact rearrangement, not an approximation —

$$F(x) = \frac{WD}{r_{\text{nat}}^2} \left[\frac{\chi(1-\rho)}{\rho^4} + \frac{(1-\nu)\chi}{\lambda^4} \left(2\rho + \frac{\chi^2(3\rho^2 + \chi^2)}{\rho^3} \right) \right]. \quad (86)$$

Once $\rho(x)$ is known the force does not depend on the membrane stiffness at all; Γ enters only through how far it lets the coil tighten. Since $\rho > \rho_0$ always and the leading bracket term falls steeply with ρ , the extensible force is always *below* the inextensible one — stretching is a relief mechanism, as expected.

Setting $\lambda \rightarrow 1$, $\rho \rightarrow \cos \phi$ and $\chi \rightarrow \sin \phi$ gives the inextensible closed form, which is the practical

one:

$$F_{\text{bend}}(x) = \frac{WD}{r_{\text{nat}}^2} \left[\frac{\sin \phi (1 - \cos \phi)}{\cos^4 \phi} + (1 - \nu) \frac{2 \sin \phi \cos^2 \phi + \sin^3 \phi}{\cos^3 \phi} \right], \quad \phi = \arcsin \frac{x}{L}. \quad (87)$$

This agrees with Eq. (86) identically, via the trigonometric identity $2 \cos^4 \phi + 3 \sin^2 \phi \cos^2 \phi + \sin^4 \phi = 2 \cos^2 \phi + \sin^2 \phi$, which is a useful independent check on the whole envelope argument. Its two limits are

$$F_{\text{bend}} \approx \frac{2(1 - \nu) WD}{r_{\text{nat}}^2} \frac{x}{L} \quad (x \ll L), \quad F_{\text{bend}} \approx \frac{WD}{4 r_{\text{nat}}^2 (1 - x/L)^2} \quad (x \rightarrow L). \quad (88)$$

The first is linear and soft, and — per the remark closing Section 6.4 — is driven entirely by the twist mismatch, the hoop term contributing only at $O(\chi^4)$. The second diverges as $(1 - x/L)^{-2}$, steeply enough to pass for an exponential over any limited fitting window.

8 Coupling II: the friction subsystem

8.1 Assembling the equations

Substituting the friction law (49)–(50) into the thickness-averaged in-plane equilibrium equations (23)–(24), and simplifying the curvature terms with $\kappa_{uu} \sin 2\phi - \kappa_{uv} \cos 2\phi = \kappa_{uv}$ (a double-angle identity applied to Eq. 10), gives the friction body force of Eq. (36) explicitly:

$$f_u^{\text{b}} = \mu \sin 2\phi \left[\frac{\Delta \sigma_{rr}}{t} + \frac{\bar{\sigma}_{rr}}{2r} \right], \quad f_v^{\text{b}} = \mu \cos 2\phi \left[\frac{\Delta \sigma_{rr}}{t} - \frac{\bar{\sigma}_{rr}}{2r} \right]. \quad (89)$$

Note that $\Delta \sigma_{rr}/t$ contains no small denominator in practice: by Eq. (60) it equals $\kappa_{uu} \bar{\sigma}_{uu} - 2\kappa_{uv} \bar{\sigma}_{uv} + \kappa_{vv} \bar{\sigma}_{vv}$ plus an $O(t)$ correction.

Counting equations and unknowns. There are five unknown fields of (u, v) : $\bar{\sigma}_{uu}$, $\bar{\sigma}_{uv}$, $\bar{\sigma}_{vv}$ and σ_{rr}^+ , σ_{rr}^- (equivalently $\bar{\sigma}_{rr}$ and $\Delta \sigma_{rr}$). It is tempting to count every relation derived so far and declare closure once the totals match, but that overstates what each buys. The radial equilibrium equation (60) and the contact recursion (53) are both statements about $\Delta \sigma_{rr}$ in terms of $\bar{\sigma}_{rr}$; together they are exactly two equations for the two quantities $\bar{\sigma}_{rr}$ and $\Delta \sigma_{rr}$, and once the in-plane stresses are known that pair is self-contained. It supplies the in-plane problem with a source term but no additional constraint. So the real gap is the one plane-stress elasticity always has: the two in-plane equilibrium equations (36) relate three unknowns, leaving one relation missing. Compatibility (34) is that relation. Properly accounted, the closed system is

two in-plane equilibrium + one compatibility	for $\bar{\sigma}_{uu}, \bar{\sigma}_{uv}, \bar{\sigma}_{vv}$
one radial equilibrium + one contact recursion	for $\sigma_{rr}^+, \sigma_{rr}^-$

— five equations for five unknowns, but arranged as two coupled subsystems rather than five constraints on everything at once.

Boundary conditions. The long edges are traction-free, so $\bar{\sigma}_{uv} = \bar{\sigma}_{vv} = 0$ at $v = \pm W/2$. Since $\bar{\sigma}_{vr}$ is tied directly to $\bar{\sigma}_{rr}$ by Eq. (49), a traction-free edge also forces $\sigma_{rr}^\pm = 0$ there individually, hence $\bar{\sigma}_{rr} = \Delta\sigma_{rr} = 0$ at the edges. Note that $\bar{\sigma}_{uu}$ is *not* constrained at a free v -edge: the traction on a face with normal $\hat{v}$ involves σ_{vv} , σ_{uv} and σ_{vr} only. At the active end, $\sigma_{rr}^+ = 0$ at $u = L$, already supplied geometrically by Section 5.4. The resultant transmitted through the cross-section at $u = L$, projected onto the pulling direction, is the applied force:

$$\boxed{F(x) = W t [\bar{\sigma}_{uu}(L) \sin \phi + \bar{\sigma}_{uv}(L) \cos \phi]}, \quad (90)$$

the factor Wt being the cross-sectional area and the bracket the traction on a face of normal $\hat{u}$ resolved along z , using $\hat{u} \cdot \hat{z} = \sin \phi$ and $\hat{v} \cdot \hat{z} = \cos \phi$ from Eq. (9). The opposite reaction force is applied at the other end, $u = 0$.

8.2 The Beltrami–Michell form

Eq. (34) still contains a mixed derivative of $\bar{\sigma}_{uv}$, which we would rather not carry. The two in-plane equilibrium equations (36) already relate the first derivatives of $\bar{\sigma}_{uv}$ to the other fields: differentiating the first with respect to u , the second with respect to v , and adding,

$$2 \frac{\partial^2 \bar{\sigma}_{uv}}{\partial u \partial v} = - \left(\frac{\partial^2 \bar{\sigma}_{uu}}{\partial u^2} + \frac{\partial^2 \bar{\sigma}_{vv}}{\partial v^2} \right) + \frac{\partial f_u^b}{\partial u} + \frac{\partial f_v^b}{\partial v}. \quad (91)$$

Substituting into Eq. (34), the coefficients of $\frac{\partial^2 \bar{\sigma}_{uu}}{\partial u^2}$ and $\frac{\partial^2 \bar{\sigma}_{vv}}{\partial v^2}$ both collect to $-\nu + (1 + \nu) = 1$, matching the coefficient already on the other two second derivatives, so the four combine into a plain Laplacian:

$$\boxed{\nabla^2 (\bar{\sigma}_{uu} + \bar{\sigma}_{vv}) = (1 + \nu) \left(\frac{\partial f_u^b}{\partial u} + \frac{\partial f_v^b}{\partial v} \right)}. \quad (92)$$

Intuitively this is a Poisson equation: the total in-plane stress cannot vary however it likes, but must curve in response to how unevenly the friction forcing is distributed, the way a temperature field curves in response to a heat source.

Eq. (92) is worth recognizing: it is exactly the standard plane-stress Beltrami–Michell equation $\nabla^2 \sigma_{kk} = -(1 + \nu) \frac{\partial b_k}{\partial x_k}$ for a body force $b_i = -f_i^b$. That identification is the promised verification that the displacement formulation of Section 4.3 solves the same problem: Eqs. (37)–(38) and Eqs. (36), (92) are two presentations of one system, and the displacement form is preferable only because it satisfies compatibility identically instead of imposing it as an extra second-order equation.

8.3 Normalization: what sets the magnitude

Every equation of this subsystem is linear and *homogeneous*. The friction body force (89) is proportional to the contact pressure, the contact pressure is generated by the in-plane stress through Eq. (60), and the traction-free conditions are homogeneous too. The trivial solution — zero stress everywhere — therefore satisfies the entire subsystem at every x . Something outside it must fix the scale, and it is worth being precise about what.

The resolution is the arc-length constraint, Eq. (2), which so far has been used only to determine ρ . It states that the ribbon's material length and its length in space must agree, i.e. that the mean axial strain is not free but equal to $\lambda - 1$:

$$\boxed{\frac{1}{LW} \int_0^L \int_{-W/2}^{W/2} \bar{\epsilon}_{uu} dv du = \frac{1}{LW} \int_0^L \int_{-W/2}^{W/2} \frac{\bar{\sigma}_{uu} - \nu \bar{\sigma}_{vv}}{E} dv du = \lambda(\rho(x)) - 1.} \quad (93)$$

This single scalar relation is inhomogeneous — its right-hand side is nonzero for every $x > 0$, by the strict inequality $\lambda > 1$ established in Section 7.2 — so it excludes the trivial solution and fixes the amplitude. In the displacement formulation it takes the especially clean form of a Dirichlet condition on the ends,

$$a(L, v) - a(0, v) = L(\lambda - 1), \quad (94)$$

which is how Section 9 imposes it. The picture is that the governing equations determine the *shape* of the stress field, and the ribbon's own inextensibility — almost, but not quite, exact — determines its *amplitude*.

Two honest caveats. First, this ties the friction field's magnitude to the geometric stretch, and those are not obviously the same physical quantity: a self-equilibrated friction pre-stress and an imposed elongation are different things, and Eq. (93) couples them. Second, the amplitude so obtained is small, since $\lambda - 1 = O(1/\Gamma)$, and it must then be multiplied by whatever amplification the friction loop provides (Section 8.6) — a product of a very small number and a very large one, which is numerically delicate. Section 9.3 quantifies how delicate.

8.4 From the residual strain to the measured force

The residual strain established in Section 7.2 is what seeds the friction field. It is worth following that chain explicitly, because doing so also settles what “the” force is.

There is one measurable force: the resultant Eq. (90) transmitted through the cross-section the Instron grips. Eq. (85) is not a rival value for it but a *decomposition* of it, and the friction subsystem modifies exactly one of the two terms. To see this, read the second term of Eq. (85) again: it is $EtW\epsilon\chi/\lambda$, which is precisely Eq. (90) evaluated on a *uniform* strain field. With $\bar{\sigma}_{uu} = E\epsilon$ everywhere and $\bar{\sigma}_{uv} = 0$,

$$Wt\bar{\sigma}_{uu}(L)\sin\phi = WtE\epsilon\frac{\chi}{\lambda} \quad (95)$$

identically. Friction does not add a new term to the force; it makes $\bar{\sigma}_{uu}$ non-uniform, concentrating it towards the ends, so that its value at $u = L$ exceeds the uniform-strain value. Naming that ratio the capstan gain,

$$\boxed{\mathcal{A}(x) := \frac{\bar{\sigma}_{uu}(L)}{E\epsilon(x)}, \quad \mathcal{A} \rightarrow 1 \text{ as } \mu \rightarrow 0,} \quad (96)$$

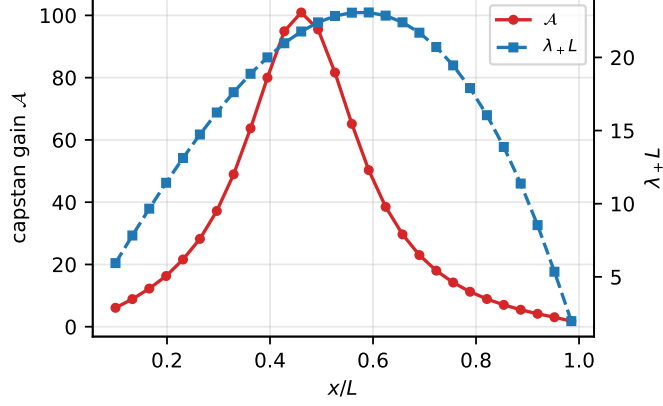


Figure 4: The capstan gain $\mathcal{A}$ of Eq. (96), measured from the two-dimensional solution of Section 9, against the accumulation exponent $\lambda_+ L$ of Eq. (103) that drives it. The gain is what the friction loop multiplies the uniform-strain membrane stress by, and it tracks the exponent as expected.

the entire force law collapses into a single expression:

$$F(x) = \underbrace{\frac{2(1-\nu)WD}{r_{\text{nat}}^2} \frac{\rho\chi}{\lambda^4}}_{\text{twist, friction-independent}} + \underbrace{\mathcal{A}(x) E t W \epsilon(x) \frac{\chi}{\lambda}}_{\text{membrane, capstan-amplified}}. \quad (97)$$

Setting $\mathcal{A} = 1$ recovers Eq. (85) exactly, and hence Eq. (87) in the inextensible limit. The twist term is untouched by friction, for two reasons already established: bending does not couple to the contact pressure (Section 6.2), and $\rho(x)$ is fixed by the friction-free balance of Section 7.2.

The causal chain from geometry to force is therefore

$$\underbrace{\lambda > 1}_{\text{Eq. (67)}} \longrightarrow \underbrace{\epsilon > 0}_{\text{Eq. (73)}} \longrightarrow \underbrace{E\epsilon}_{\text{uniform membrane stress}} \longrightarrow \underbrace{\mathcal{A}E\epsilon}_{\text{friction amplification}} \longrightarrow \underbrace{F}_{\text{Eq. (97)}}, \quad (98)$$

and no step in it is adjustable. Every arrow is an equation already derived. The first arrow is the one that carries the weight: were $\lambda = 1$ exactly, every subsequent quantity would vanish and the model would predict no friction force at all. The strict inequality $\lambda > 1$ of Section 7.2 is what makes the friction branch exist.

The gain itself is measured directly from the two-dimensional solution (Section 9.2) and is plotted in Figure 4. It runs from about 2 to about 100, peaking near $x/L \approx 0.45$, and tracks the accumulation exponent $\lambda_+ L$ of Eq. (103) as it should — the gain is what that exponent buys once the boundary conditions have taken their cut. It is also why the two terms of Eq. (97) trade places: the twist term dominates below $x/L \approx 0.2$, and the amplified membrane term above it.

8.5 Why this is not an energy minimization

It is tempting to treat the whole coupled problem the way Section 7 treated the radius: write the total stored energy including the friction-driven membrane stress, and minimize. This is wrong,

and the reason is worth recording because the error is not obvious.

Minimization is legitimate only for a recoverable potential. The friction-amplified membrane stress does store elastic energy, but that energy is held in place by a dissipative, history-dependent contact law: release the load and the stress does not simply run back down the same path. In a rate-independent quasi-static model there is no way to separate the stored-and-recoverable part from the stored-but-friction-locked part, so the total membrane energy is not an admissible potential for a variational argument.

The consequence is quantitative rather than merely philosophical. Feeding the solved field's strain non-uniformity $\Xi := \langle \bar{\epsilon}_{uu}^2 \rangle / \langle \bar{\epsilon}_{uu} \rangle^2$ back into Eq. (65) as an effective membrane stiffening gives $\Xi \sim 4 \times 10^5$ for the parameters of Section 9.2, which drives the minimizer of Eq. (67) to the inextensible radius, collapses ϵ to about 10^{-12} , and collapses F with it. Only the bare, uniform-strain membrane stiffness of Eq. (64) belongs in the radius balance.

The same objection disqualifies a tempting cross-check. One might try to obtain F a second way, as $\frac{dU_{\text{elastic}}}{dx}$ plus a Coulomb dissipation integral, and compare it against Eq. (97). That is not an independent check but an inadmissible one: it requires splitting the stored energy into recoverable and friction-locked parts, which is precisely what a rate-independent model cannot do. Carried out anyway, it disagrees with Eq. (97) by factors of two to five — a symptom of the illegitimacy, not evidence of an error in either. Eq. (97), one force decomposed into two additive terms, is the statement to use.

8.6 Width-averaged reduction and the accumulation exponent

Before solving the full two-dimensional system it is worth extracting the essential behaviour analytically, because it explains both the shape of the answer and the numerical difficulty.

Integrate the in-plane $\hat{u}$ equation (36) across the width and divide by W . The $\frac{\partial \bar{\sigma}_{uv}}{\partial v}$ term integrates exactly to its edge values, which vanish, so with $\hat{\sigma}_{uu}(u)$ and $\hat{\sigma}_{rr}(u)$ denoting the width averages of $\bar{\sigma}_{uu}$ and $\bar{\sigma}_{rr}$, and using Eq. (60) at leading order for $\Delta \sigma_{rr}$,

$$\frac{d\hat{\sigma}_{uu}}{du} = \mu \sin 2\phi \left[\kappa_{uu} \hat{\sigma}_{uu} + \frac{\hat{\sigma}_{rr}}{2r} \right]. \quad (99)$$

Doing the same to the transport equation (53), whose $\frac{\partial \bar{\sigma}_{rr}}{\partial v}$ term likewise integrates to vanishing edge values,

$$\frac{d\hat{\sigma}_{rr}}{du} = \frac{t \kappa_{uu}}{2\pi r \cos \phi} \hat{\sigma}_{uu}. \quad (100)$$

Both steps are exact consequences of the edge conditions rather than mode truncations. This is a linear, constant-coefficient 2×2 system in u , the coefficients depending on x only: $\frac{d}{du} (\hat{\sigma}_{uu}, \hat{\sigma}_{rr})^\top = \begin{pmatrix} A & B \\ C & 0 \end{pmatrix} (\hat{\sigma}_{uu}, \hat{\sigma}_{rr})^\top$ with

$$A = \mu \sin 2\phi \kappa_{uu}, \quad B = \frac{\mu \sin 2\phi}{2r}, \quad C = \frac{t \kappa_{uu}}{2\pi r \cos \phi}. \quad (101)$$

Its trace is A and its determinant $-BC < 0$, so the eigenvalues

$$\lambda_{\pm} = \frac{A \pm \sqrt{A^2 + 4BC}}{2} \quad (102)$$

are always *real*, with $\lambda_+ > 0 > \lambda_-$. The system is a saddle, never a rotation, so $\hat{\sigma}_{uu}$ and $\hat{\sigma}_{rr}$ are sums of real exponentials in u and never oscillatory — a useful check, since a mode-truncated treatment of the same equations can produce spurious oscillation by coupling a mode to a frequency the edge conditions do not admit. Since $C \propto t$ is small, $\lambda_+ \approx A$ and the growing mode is

$$\boxed{\lambda_+ \approx \mu \sin 2\phi \kappa_{uu} = \frac{\mu \sin 2\phi \cos^2 \phi}{r} \implies \lambda_+ L \approx 0.77 \mu \theta_L.} \quad (103)$$

The last step uses $r = r_{\text{nat}} \cos \phi$ and $L/r_{\text{nat}} = \theta_L$, then maximizes $\sin 2\phi \cos \phi \leq 0.77$ over ϕ . This is the single most consequential number in the friction subsystem, so it is worth reading carefully: the total amplification exponent along the ribbon is set by the friction coefficient times the *number of turns*, and is completely independent of E , t and W . For $\mu = 0.3$ and thirty turns it is 46, an amplification of $e^{46} \sim 10^{20}$ between the two ends of the ribbon. This is the capstan effect, and it is why the friction contribution is at once physically dominant and numerically treacherous.

9 Numerical solution

9.1 Formulation

The full two-dimensional system is solved on a finite-difference grid in (u, v) at each frozen x . Four choices are worth recording, each made to avoid a specific failure.

1. **Displacement unknowns.** The primary fields are a and b of Eq. (35), solved through Eqs. (37)–(38), rather than the three stresses through Eqs. (36) and (92). Compatibility is then identical rather than imposed, which makes the assembled system square and consistent; and the friction body force appears undifferentiated, so nothing interpolated is ever differentiated.
2. **Exact contact recursion.** σ_{rr}^+ is carried as a third unknown field satisfying Eqs. (51)–(52) directly, with bilinear interpolation at the generally off-grid contact partner, rather than the Taylor-expanded transport equation (53). This enforces $\sigma_{rr}^+(L, v) = 0$ geometrically and avoids finite-differencing a first-order equation across the narrow bands where both neighbours are present.
3. **Elongation as a Dirichlet condition.** Eq. (94) supplies the normalization, so no arbitrary scale anchor is needed anywhere. Laterally, $u = 0$ is fully clamped ($b = 0$) and $u = L$ is left shear-free ($\bar{\sigma}_{uv} = 0$). Clamping both ends laterally places a clamped-meets-free corner singularity at $u = L$, exactly where Eq. (90) is evaluated; imposing $\bar{\sigma}_{uv} = 0$ at both ends instead over-constrains the global axial balance, because the equilibrium equations are written in the rotating local frame, so $\iint f_u^b du dv$ is not a conserved quantity.

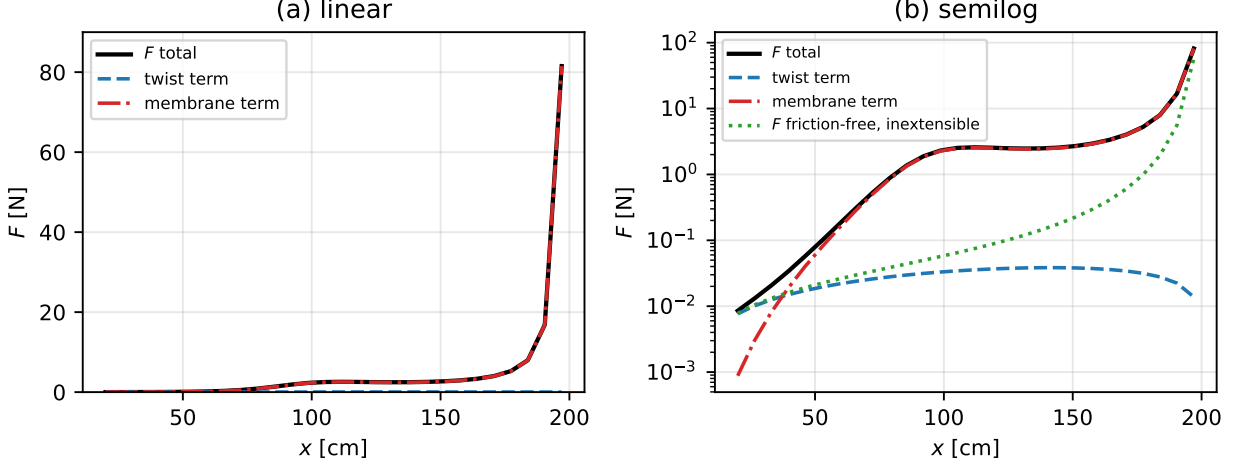


Figure 5: Force–displacement, linear and semilogarithmic, showing the two additive terms of Eq. (97). The twist term is friction-independent and dominates below $x/L \approx 0.2$; the capstan-amplified membrane term dominates above it. The dotted curve is the friction-free closed form Eq. (87) for comparison, which the total exceeds once the gain $\mathcal{A}$ of Eq. (96) departs from unity. The total spans 0.0085 to 82 N with a logarithmic slope that *increases* with x , i.e. growth faster than any single exponential.

4. **Resolution set by turns, not by length.** The contact partner sits $2\pi r \cos \phi$ away in u , so the recursion is meaningless unless several grid points fit inside that distance. Measured convergence: F is converged to a few percent at ten or more points per turn, and not converged at all below three. Since the offset shrinks as the coil tightens, the grid is adapted per x , reaching several thousand points along u near full extension.

9.2 Results

Figures 3–6 use $L = 200$ cm, $W = 8$ cm, $t = 0.01$ cm, $r_{\text{nat}} = 2$ cm (hence $\theta_L = 100$ rad, about 15.9 turns), $E = 3 \times 10^5$ N/cm², $\nu = 0.3$ and $\mu = 0.3$. These give $\Gamma = 4.4 \times 10^5$, $\rho_* = 0.13$ and $\rho_{\text{jam}} = 0.040$ — so the elastic floor of Eq. (78) is the operative one — and a peak amplification exponent $\lambda_+ L \approx 23$.

Three results stand out.

First, F_{bend} is strictly monotonic and grows faster than exponentially: its logarithmic slope $\frac{d(\ln F)}{dx}$ rises from 0.020 to 0.36 cm⁻¹ across the sweep. It spans four orders of magnitude, 0.0078 to 58 N, with no fitted parameter anywhere. The closed form (87) and the numerically differentiated energy agree to within a couple of percent (Figure 5), which validates the envelope argument of Section 7.4 and the numerical energy bookkeeping against each other.

Second, the total stored elastic energy is overwhelmingly bending, not membrane: $U_{\text{mem}}/U_{\text{bend}}$ runs from 10^{-4} to 10^{-3} over almost the whole range, rising to 0.4 only in the last few percent of extension. So $\frac{dU_{\text{elastic}}}{dx}$ is, to a fraction of a percent, the pure bending force of Eq. (87). The elastic branch of the model is essentially entirely Section 6’s physics.

Third, the capstan-amplified membrane term of Eq. (97) overtakes the twist term at $x/L \approx 0.2$

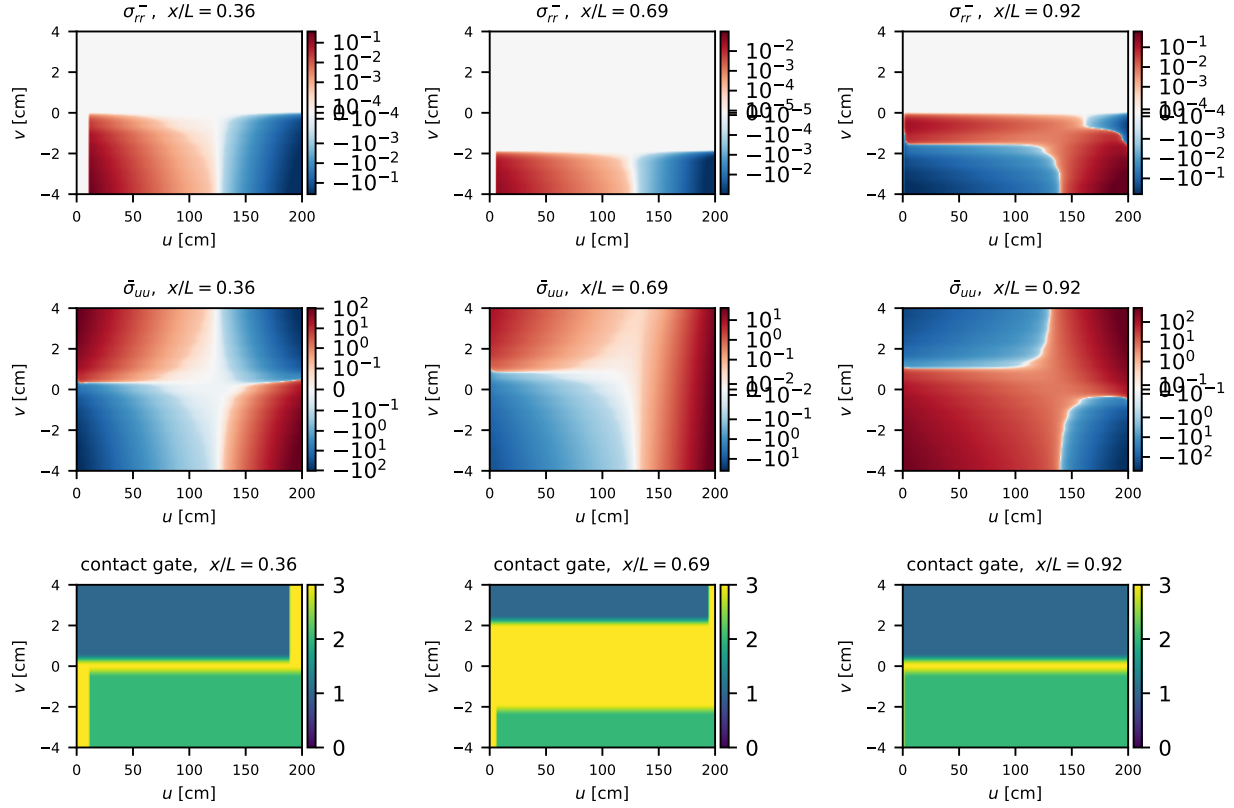


Figure 6: Stress fields at three extensions. Top: contact pressure σ_{rr}^- . Middle: in-plane $\bar{\sigma}_{uu}$, showing the boundary-layer structure implied by the amplification exponent Eq. (103) — stress concentrated within $1/\lambda_+$ of the ends, with a nearly stress-free interior. Bottom: the contact gates of Eqs. (40)–(41), sweeping across the width as the axial offset between turns grows.

and dominates thereafter, the measured gain $\mathcal{A}$ of Eq. (96) reaching about 100 near $x/L \approx 0.45$ (Figure 4). The total force shows a shallow non-monotonic plateau between $x/L \approx 0.53$ and 0.72. That plateau is not numerical noise: it tracks the contact-area collapse predicted in Section 5.1, the fraction of the sheet possessing an inner neighbour falling from 0.79 to 0.22 and recovering to 0.69 across the sweep as the axial offset $2\pi r \sin \phi$ passes through its maximum. Its depth is governed by $W/(\pi r_{\text{nat}})$, which is 1.27 here; at 2.04 the contact fraction only falls to 0.52 and the plateau largely disappears.

9.3 Limits of the numerical treatment

These are stated in full because they bound what the friction half of the model can currently be said to predict.

A conditioning wall. By Eq. (103) the amplification exponent is $\lambda_+ L \approx 0.77 \mu \theta_L$. At thirty turns and $\mu = 0.3$ it peaks near 46, giving a ratio between the growing and decaying modes of $e^{46} \sim 10^{20}$, against the $\sim 10^{16}$ that double precision carries. The two-point boundary value problem in u is then genuinely unsolvable: at $x/L = 0.6$ with those parameters, successively refined grids

give $F = 14.5, 41.1, -80.2, -26.3$ N — sign-flipping noise, not a converging sequence. This is a conditioning limit rather than a discretization error, and grid refinement does not touch it. The physically correct treatment in that regime is a one-sided boundary-layer problem near $u = L$ rather than a two-point problem across the whole ribbon; that has not been carried out. The parameters quoted in Section 9.2 keep $\lambda_+ L \approx 23$, which is solvable and converges.

A parameter squeeze. Two requirements pull against each other. Self-contact needs πr_{nat} comfortably below W (Section 5.1), which favours a small coil radius; conditioning needs $\mu L/r_{\text{nat}}$ small, which favours a large one. At $L = 200$ cm and $W = 8$ cm both can be met, but only with modest margin.

The energy cross-check is inadmissible. There is only one force, Eq. (97), and it is what the solver reports. The alternative of computing $\frac{dU_{\text{elastic}}}{dx}$ plus a Coulomb dissipation integral is not a second route to the same number but an inadmissible one, for the reason given in Section 8.5: it presumes the stored energy can be split into recoverable and friction-locked parts. Evaluated regardless, it differs by factors of two to five. This is worth stating only so that the discrepancy is not mistaken for a numerical failure.

Unilateral contact is not enforced. Assumption 7 of Section 1.2 requires $\sigma_{rr} \leq 0$, but the governing equations contain no complementarity condition, and the solution does return tensile patches — up to about half the contact area at large extension. Doing this properly is a free-boundary problem in which contact is lost wherever the pressure would go positive, and it is not solved here. Since the friction magnitude depends on $|\sigma_{rr}|$, this is a genuine quantitative caveat on the friction branch, though it does not touch F_{bend} .

10 Predictions and experimental validation

The two branches of the model make predictions of very different quality, and the difference is worth stating plainly.

F_{bend} (Eq. 87) contains no adjustable parameter. It is monotonic, faster than exponential, and its magnitude is set by WD/r_{nat}^2 . For the geometry of Section 9.2 the soft-branch slope of Eq. (88) evaluates to about 0.19 N, the same order as the prefactor of the slowly growing term in the measured fit of Figure 7. That is encouraging but not decisive, since a curve of roughly the right shape and magnitude can arise for more than one reason.

Two sharper tests follow from F_{bend} ’s functional form, and both discriminate against a friction-dominated reading:

- **Independence of μ .** F_{bend} contains no friction coefficient at all. Changing the interface — a different paper stock, talc, graphite, a change in humidity — should leave $F(x)$ essentially unchanged if the bending branch dominates, and change it substantially otherwise.
- **Cubic thickness scaling.** $F_{\text{bend}} \propto D \propto t^3$, and also $\propto W$ and $\propto r_{\text{nat}}^{-2}$. Doubling the sheet thickness at fixed winding geometry should multiply the force by eight. A cubic law is hard

to confuse with anything else, and this is probably the single most decisive measurement available.

The friction branch, by contrast, predicts a force proportional to μ and amplified by $e^{0.77\mu\theta L}$, so it should be extremely sensitive to both the interface and the number of turns. The number of turns is easy to vary and is a clean second discriminator.

Finally, Section 7.3 predicts a sliding-to-stretching crossover at $1 - x_*/L \approx 2^{-4/3}\Gamma^{-1/3} \sim (t/r_{\text{nat}})^{2/3}$, with a crossover radius $\rho_* \approx (4/\Gamma)^{1/6}$ that is independent of the material. Two things make this testable. The crossover displacement and radius should both be unchanged by a change of sheet stiffness at fixed geometry, since Γ does not contain E . And the crossover should move with the winding as Eq. (83) dictates: for a coil wound with enough turns that $\rho_{\text{jam}}/\rho_* > 1$, the radius scale should instead track half the measured radial build over the initial radius. Measuring the coil radius where the sliding visibly slows, across two windings on opposite sides of the contour in Figure 3(c), would test Eq. (78) and Eq. (83) together. What should *not* be looked for is a sharp stop: Eq. (76) makes the transition a gradual reallocation, and the coil keeps tightening slowly well past x_* .

11 The elastic energy in closed form

Section 7 settled the coil radius by minimizing an energy built from two scalars, a bending density and a membrane density $\frac{1}{2}E\epsilon^2$. This section derives that energy instead of assuming it, starting from the general elastic energy density and eliminating unknowns one at a time until nothing is left but the geometry.

The destination is a closed-form $U(\rho; \chi)$. The route is a ledger: at each step either an unknown field is removed, or an equation is added, and every removal is charged to a numbered assumption of Section 1.2 rather than to the smallness of a particular number. Table 1 keeps the count, and the three equilibrium equations are used rather than dropped: the $\hat{u}$ and $\hat{v}$ pair in Sections 11.6–11.8, the radial one in Section 11.9. Several steps do not come out cleanly, and each carries a **flagged note** naming what depends on it.

11.1 Twelve unknowns

At a material point (u, v, ζ) the stored elastic energy per unit volume is

$$u_{\text{el}} = \frac{1}{2} \sigma_{ij} \epsilon_{ij} = \frac{1}{2} \left[\sigma_{uu} \epsilon_{uu} + \sigma_{vv} \epsilon_{vv} + \sigma_{rr} \epsilon_{rr} + 2\sigma_{uv} \epsilon_{uv} + 2\sigma_{ur} \epsilon_{ur} + 2\sigma_{vr} \epsilon_{vr} \right]. \quad (104)$$

Nothing has been assumed yet. The stress carries the six independent components of Eq. (14) and the strain carries six of its own, so Eq. (104) contains *twelve* unknown fields of three variables. Equilibrium (Section 3) supplies three equations and the constitutive law (Section 11.4 below) supplies six, which is not enough on its own; the rest must come from kinematics, from the contact recursion, and from the boundary conditions. That is the accounting problem this section solves.

11.2 Normality: two strains go, and two stresses stop working

Assumption 8 states that material normals to the mid-surface stay normal and stay unstretched, which is precisely the statement

$$\boxed{\epsilon_{ur} = \epsilon_{vr} = 0.} \quad (105)$$

Two unknowns are gone, leaving ten. Note carefully what has *not* been assumed: σ_{ur} and σ_{vr} are not zero. They are the reactions that enforce Eq. (105), exactly as the transverse shear force in an Euler–Bernoulli beam is a reaction rather than a constitutive response, and they will be needed in equilibrium.

Their one consequence for the energy is decisive. In Eq. (104) they appear only against their conjugate strains, so

$$\boxed{\sigma_{ur}\epsilon_{ur} + \sigma_{vr}\epsilon_{vr} = 0 \quad \text{identically: the transverse shears are } \textit{workless}.} \quad (106)$$

Two things follow immediately. Friction acts on the ribbon exclusively through σ_{ur} and σ_{vr} (Eqs. 47–48), so friction contributes nothing to the stored elastic energy — which is the exact sense in which Section 8.5 is right that friction cannot be folded into a potential, and the reason a friction-free treatment of the energy costs nothing. And the energy needs only eight of the ten survivors: $\sigma_{uu}, \sigma_{vv}, \sigma_{uv}, \sigma_{rr}$ and $\epsilon_{uu}, \epsilon_{vv}, \epsilon_{uv}, \epsilon_{rr}$.

11.3 Linearizing across the thickness

The ten survivors are still fields of (u, v, ζ) . This step removes the ζ dependence, converting them into fields of (u, v) .

For the in-plane strains the reduction is exact and free. Normality plus a mid-surface displacement field forces the strain to vary affinely through the thickness: a fibre at offset ζ lies at radius $r + \zeta$ where its natural state put it at $r_{\text{nat}} + \zeta$, so its length ratio relative to the mid-surface is $(1 + \zeta/r)(1 - \zeta/r_{\text{nat}})$, and to first order in ζ

$$\boxed{\epsilon_{uu} = \bar{\epsilon}_{uu} + \zeta \Delta\kappa_{uu}, \quad \epsilon_{vv} = \bar{\epsilon}_{vv} + \zeta \Delta\kappa_{vv}, \quad \epsilon_{uv} = \bar{\epsilon}_{uv} + \zeta \Delta\kappa_{uv}.} \quad (107)$$

The gradients $\Delta\kappa_{ij}$ are not unknowns: they are the curvature mismatch of Eq. (54), fixed by the geometry, and they depend on x alone. So each in-plane strain collapses to a single unknown function of (u, v) , its mid-surface value, which by Eq. (15) is also its thickness average. Three unknowns remain three, but they are now functions of two variables instead of three.

For the contact pressure the same step *costs* an unknown. Nothing in the kinematics constrains σ_{rr} through the thickness, so it is assumption 5 that supplies the two surface coefficients:

$$\boxed{\sigma_{rr}(\zeta) = \bar{\sigma}_{rr} + \frac{\zeta}{t} \Delta\sigma_{rr} + \frac{P}{2} \left(\zeta^2 - \frac{t^2}{4} \right).} \quad (108)$$

Why the quadratic term, and why it is not a third unknown. Radial equilibrium (22) gives $\frac{\partial \sigma_{rr}}{\partial \zeta}$ as the hoop projection of the in-plane stress. That projection is *almost* affine in ζ :

the Kirchhoff part is affine by Eq. (107), but σ_{uu} and σ_{vv} each carry $\frac{\nu}{1-\nu}\sigma_{rr}$ from Eq. (112), and $\kappa_{uu} + \kappa_{vv} = 1/r$, so the equation closes on itself:

$$\frac{\partial \sigma_{rr}}{\partial \zeta} = \underbrace{A + B\zeta}_{\text{Kirchhoff}} + g\sigma_{rr}, \quad g := \frac{\nu}{(1-\nu)r}. \quad (109)$$

This is a linear first-order ODE, whose exact solution is an affine particular part plus $C e^{g\zeta}$ — so σ_{rr} is strictly exponential, on the length scale $(1-\nu)r/\nu \approx 2.3r$. Expanding $e^{g\zeta}$ and truncating,

$$\sigma_{rr} = [\text{constant}] + [\text{linear}]\zeta + [\text{quadratic}]\zeta^2 + O(g\zeta) \times [\text{quadratic}], \quad \frac{gt}{2} = \frac{\nu t}{2(1-\nu)r}, \quad (110)$$

so quadratic is the correct truncation and the neglected cubic is smaller by $gt/2$, which is 6.7×10^{-3} even at the tightest radius the coil reaches. That is $t \ll r$, i.e. assumption 9 again, and it is the same group that governed Eq. (115). So the quadratic term is forced by Eq. (22) rather than assumed, and the profile could not have been taken linear.

It costs no third unknown field, for two reasons. First, $\zeta^2 - t^2/4$ vanishes at *both* faces, so

$$\frac{1}{2} [\sigma_{rr}^+ + \sigma_{rr}^-] = \bar{\sigma}_{rr}, \quad \sigma_{rr}^+ - \sigma_{rr}^- = \Delta\sigma_{rr}, \quad \frac{1}{t} \int_{-t/2}^{t/2} \sigma_{rr} d\zeta = \bar{\sigma}_{rr} - \frac{Pt^2}{12}, \quad (111)$$

so $\bar{\sigma}_{rr}$ and $\Delta\sigma_{rr}$ keep exactly the meanings the recursion (53) and the friction law (49) need — surface mean and surface difference — and the quadratic term disturbs neither. This is the parity argument of Section 6.3 seen from the other side. Note, though, that $\bar{\sigma}_{rr}$ is now the surface mean and *not* the thickness average, which differ by $Pt^2/12$; the average is what enters the stress resultants, and Section 11.11 uses it there. Second, P is not free: Section 11.9 determines it from the first ζ -moment of the same equation. So one field of (u, v, ζ) becomes two unknown fields of (u, v) plus one determined by them. The same applies to the two transverse shears, giving $\bar{\sigma}_{ur}, \Delta\sigma_{ur}, \bar{\sigma}_{vr}, \Delta\sigma_{vr}$. The tally is now nine unknown functions of (u, v) : $\bar{\epsilon}_{uu}, \bar{\epsilon}_{vv}, \bar{\epsilon}_{uv}$; $\bar{\sigma}_{rr}, \Delta\sigma_{rr}$; and the four shear coefficients, of which none enters the energy.

11.4 The constitutive law, and an exact split of the energy

Because σ_{rr} does not vanish — the layers press on one another (Section 5.4) — Eq. (31) is not the right constitutive statement here. The state through the thickness is genuinely triaxial: it is plane *stress* that a free thin sheet enjoys, and even then $\epsilon_{rr} = -\nu(\sigma_{uu} + \sigma_{vv})/E \neq 0$, so ϵ_{rr} could never have been discarded either. Writing the full isotropic law and solving it for the in-plane stresses in terms of the in-plane strains and σ_{rr} ,

$$\boxed{\begin{aligned} \sigma_{uu} &= \frac{E}{1-\nu^2} (\epsilon_{uu} + \nu \epsilon_{vv}) + \frac{\nu}{1-\nu} \sigma_{rr}, \\ \sigma_{vv} &= \frac{E}{1-\nu^2} (\epsilon_{vv} + \nu \epsilon_{uu}) + \frac{\nu}{1-\nu} \sigma_{rr}, \end{aligned} \quad \sigma_{uv} = \frac{E}{1+\nu} \epsilon_{uv},} \quad (112)$$

which reduces to Eq. (32) when $\sigma_{rr} = 0$, together with

$$\epsilon_{rr} = \frac{(1+\nu)(1-2\nu)}{(1-\nu)E} \sigma_{rr} - \frac{\nu}{1-\nu} (\epsilon_{uu} + \epsilon_{vv}). \quad (113)$$

The coefficient in the first term is the reciprocal of the confined modulus: the stiffness the material shows when squeezed while forbidden to spread. The second term is ordinary Poisson thinning under in-plane tension.

Four unknowns are eliminated. $\sigma_{uu}, \sigma_{vv}, \sigma_{uv}$ and ϵ_{rr} are now expressions in $\epsilon_{uu}, \epsilon_{vv}, \epsilon_{uv}, \sigma_{rr}$, and since those four are affine in ζ by Eqs. (107) and (108), so are the four outputs — the linearization propagates through the constitutive law rather than having to be assumed again. Five unknown functions of (u, v) remain in the energy's chain: $\bar{\epsilon}_{uu}, \bar{\epsilon}_{vv}, \bar{\epsilon}_{uv}, \bar{\sigma}_{rr}, \Delta\sigma_{rr}$.

Substituting Eqs. (112)–(113) into Eq. (104) is worth doing carefully, because the terms in $\sigma_{rr}(\epsilon_{uu} + \epsilon_{vv})$ cancel exactly between $\frac{1}{2}(\sigma_{uu}\epsilon_{uu} + \sigma_{vv}\epsilon_{vv})$ and $\frac{1}{2}\sigma_{rr}\epsilon_{rr}$, leaving

$$u_{\text{el}} = \underbrace{\frac{E}{2(1-\nu^2)} (\epsilon_{uu}^2 + 2\nu\epsilon_{uu}\epsilon_{vv} + \epsilon_{vv}^2)}_{\text{in-plane}} + \underbrace{\frac{E}{1+\nu} \epsilon_{uv}^2 + \frac{(1+\nu)(1-2\nu)}{2(1-\nu)E} \sigma_{rr}^2}_{\text{through-thickness}}. \quad (114)$$

This is exact, not an expansion, and it is the first real simplification of the energy: parameterized by the in-plane strains and σ_{rr} , the two mechanisms do not couple in the density, and the transverse term is strictly positive and quadratic in the pressure. Squeezing the stack always costs energy and never pays. The coupling has not vanished — it has moved into the boundary conditions, and Section 11.11 is where it reappears.

11.5 The through-thickness measure

One check before any ζ integral: the element $du dv d\zeta$ is not quite the volume. On a surface whose principal curvatures sum to $\kappa_{uu} + \kappa_{vv}$ with Gaussian curvature K , the offset surface at ζ carries area element $1 + \zeta(\kappa_{uu} + \kappa_{vv}) + \zeta^2 K$. Here $K = 0$ exactly (Eq. 12) and the trace is $1/r$ (Eq. 55), so the quadratic term drops and

$$dV = \left(1 + \frac{\zeta}{r}\right) du dv d\zeta, \quad (115)$$

with no higher-order terms at all, while a face of normal $\hat{v}$ carries $(1 + \zeta\kappa_{uu}) du d\zeta$ and a face of normal $\hat{u}$ carries $(1 + \zeta\kappa_{vv}) dv d\zeta$. Assumption 9 sets all three weights to unity.

Two remarks on that. It is not an independent approximation: it is exactly what Section 3.2 already did in treating the faces of a volume element as rectangular, so retaining the weight in the energy while dropping it in the equilibrium equations would leave the two formulations of this model describing different bodies. And the right place to evaluate it is the *reference* configuration, because Eq. (114) is a linear-elastic strain-energy *density* and therefore integrates over the reference volume; using the deformed volume instead differs at relative order ϵ , which is smaller than the

weight itself.

The term this discards is worth naming, because it is the only place a thickness-averaged treatment loses something and because the bending part of Eq. (107) is odd in ζ and would otherwise integrate away. Carrying the weight through gives one surviving cross term, with every ν cancelled,

$$u_{\text{el}}^{\text{cross}} = \frac{t^2}{12r} \bar{\epsilon}_{uu} \Delta\kappa_{uu} \quad (\text{discarded by assumption 9}), \quad (116)$$

a direct coupling of membrane strain to bending mismatch, negative once the coil has tightened and $\Delta\kappa_{uu} < 0$: the extra material on the outer half of the thickness relieves a little of the bending penalty. Relative to the retained energy it is of order $\frac{t^2 \bar{\epsilon}_{uu}}{r^2} / \frac{t^2 \Delta\kappa_{uu}^2}{r_{\text{nat}}^2} \sim \epsilon$, and Section 11.13 reports the measured value.

Dropping the weight also collapses a distinction that would otherwise have to be carried: with unit weight the mid-surface strain and the thickness average coincide identically, so $\bar{\epsilon}_{ij}$ carries both meanings and the arc-length constraint may be read either way.

11.6 The moments, and the source term they supply

The moment balance of Section 3.4 has not been used yet. To use it we need the moments, and the point of this subsection is that they are not new unknowns: Eq. (107) has made every in-plane stress an affine function of ζ , so the M_{ij} of Eq. (26) are ζ -moments of a field the constitutive law has already fixed.

Substituting Eqs. (107) and (108) into the first line of Eq. (112) and sorting by parity in ζ ,

$$\sigma_{uu}(\zeta) = \underbrace{\frac{E}{1-\nu^2} (\bar{\epsilon}_{uu} + \nu \bar{\epsilon}_{vv}) + \frac{\nu}{1-\nu} \bar{\sigma}_{rr}}_{\text{even}} + \zeta \underbrace{\left[\frac{E}{1-\nu^2} (\Delta\kappa_{uu} + \nu \Delta\kappa_{vv}) + \frac{\nu}{1-\nu} \frac{\Delta\sigma_{rr}}{t} \right]}_{\text{odd}}. \quad (117)$$

Only three ζ integrals are needed, and with the unit weight of Section 11.5 they are elementary:

$$\int_{-t/2}^{t/2} d\zeta = t, \quad \int_{-t/2}^{t/2} \zeta d\zeta = 0, \quad \int_{-t/2}^{t/2} \zeta^2 d\zeta = \frac{t^3}{12}. \quad (118)$$

The middle one separates the two brackets: the even bracket is the whole of the resultant $\bar{\sigma}_{uu}$ and contributes nothing to the moment, the odd bracket the reverse. Multiplying by ζ and integrating therefore annihilates the even bracket and multiplies the odd one by $t^3/12$,

$$M_{uu} = \frac{t^3}{12} \frac{E}{1-\nu^2} (\Delta\kappa_{uu} + \nu \Delta\kappa_{vv}) + \frac{t^3}{12} \frac{\nu}{1-\nu} \frac{\Delta\sigma_{rr}}{t}, \quad (119)$$

and recognizing $\frac{Et^3}{12(1-\nu^2)} = D$ in the first term while cancelling one power of t in the second,

$$\boxed{\begin{aligned} M_{uu} &= D (\Delta\kappa_{uu} + \nu \Delta\kappa_{vv}) + \frac{\nu t^2}{12(1-\nu)} \Delta\sigma_{rr}, \\ M_{vv} &= D (\Delta\kappa_{vv} + \nu \Delta\kappa_{uu}) + \frac{\nu t^2}{12(1-\nu)} \Delta\sigma_{rr}, \\ M_{uv} &= D (1-\nu) \Delta\kappa_{uv}. \end{aligned}} \quad (120)$$

The second line follows identically from the second line of Eq. (112), which differs only by $uu \leftrightarrow vv$ and carries the same pressure term; the third follows the same way from $\sigma_{uv} = \frac{E}{1+\nu} \epsilon_{uv}$ using $\frac{Et^3}{12(1+\nu)} = D(1-\nu)$, with no pressure term because σ_{rr} does not enter the shear response. The D terms are *identically* the Kirchhoff–Love closure that Eq. (56) posited: expanding that equation,

$$D [(1-\nu) \Delta\kappa_{uu} + \nu (\Delta\kappa_{uu} + \Delta\kappa_{vv})] = D (\Delta\kappa_{uu} + \nu \Delta\kappa_{vv}), \quad (121)$$

and M_{uv} agrees term for term. So Section 6.2 is *derived* here rather than assumed, and the only genuinely new piece is the pressure moment: a pressure differing between the two faces is a bending load, because it stresses one face more than the other.

Where the moments enter equilibrium, and what that is worth. The moments act through the first ζ -moment of Eqs. (20)–(21), already taken in Section 3.4. It is worth being blunt about the accounting before doing anything with it: **this subsection removes no unknown and adds no equation.** The first moment is one linear functional of a pointwise equation that Section 11.7 goes on to solve in full, so it cannot be independent of it. What the moments are needed for is different, and concrete. Eq. (120) is the only route by which the bending part of the in-plane stress reaches the transverse-shear problem: the source in that problem is the through-thickness *slope* of the in-plane stress gradients, and by Eq. (132) that slope is exactly $\frac{12}{t^3} \nabla \cdot M$. Without M_{ij} there is no source to write down, no parabolic shear profile, and no way to see which of the two solvability conditions in Section 11.7 is exact. The moments are machinery, and this subsection assembles it.

Specializing Eqs. (29)–(30) to a shear-free interface, where the surface bracket vanishes by Eq. (130) below, leaves

$$\boxed{\begin{aligned} \frac{\partial M_{uu}}{\partial u} + \frac{\partial M_{uv}}{\partial v} + \kappa_{uu} \Xi_{ur} - \kappa_{uv} \Xi_{vr} &= Q_u, \\ \frac{\partial M_{uv}}{\partial u} + \frac{\partial M_{vv}}{\partial v} - \kappa_{uv} \Xi_{ur} + \kappa_{vv} \Xi_{vr} &= Q_v, \end{aligned}} \quad (122)$$

with Ξ_{ir} from Eq. (26) and Q_i from Eq. (27). Every quantity on the left is known once the constitutive law and the geometry are in hand except Ξ_{ur} and Ξ_{vr} , and the right-hand side is unknown outright, so as it stands Eq. (122) is two equations in four unknowns. That is the honest starting point.

Rotating into the circumferential and axial directions. The matrix acting on (Ξ_{ur}, Ξ_{vr}) in Eq. (122) is worth naming and diagonalizing, because the same matrix reappears in Section 11.7. Substituting the curvatures of Eq. (10),

$$\mathbf{K} := \begin{pmatrix} \kappa_{uu} & -\kappa_{uv} \\ -\kappa_{uv} & \kappa_{vv} \end{pmatrix} = \frac{1}{r} \begin{pmatrix} \cos^2 \phi & -\sin \phi \cos \phi \\ -\sin \phi \cos \phi & \sin^2 \phi \end{pmatrix} = \frac{1}{r} \vec{n} \vec{n}^\top, \quad \vec{n} := \begin{pmatrix} \cos \phi \\ -\sin \phi \end{pmatrix}, \quad (123)$$

the last step being nothing more than the outer product of $\vec{n}$ with itself. Here and below $\vec{n}$ and

$$\vec{m} := \begin{pmatrix} \sin \phi \\ \cos \phi \end{pmatrix} \quad (124)$$

are two-component column vectors holding the $(\hat{u}, \hat{v})$ components of $\hat{\theta}$ and $\hat{z}$: inverting Eq. (9) gives $\hat{\theta} = \cos \phi \hat{u} - \sin \phi \hat{v}$ and $\hat{z} = \sin \phi \hat{u} + \cos \phi \hat{v}$. They are orthonormal, so $\{\vec{n}, \vec{m}\}$ is a basis of the tangent plane, and Eq. (123) says $\mathbf{K}$ is $1/r$ times the projector onto the circumferential direction:

$$\boxed{\mathbf{K} \vec{n} = \frac{\vec{n}}{r}, \quad \mathbf{K} \vec{m} = \vec{0}.} \quad (125)$$

That is the entire geometric content of $K = 0$ (Eq. 12): the surface is curved in one direction and flat in the other, so $\mathbf{K}$ has one nonzero eigenvalue and one null direction, and $\det \mathbf{K} = 0$ follows.

Two directional derivatives go with the same pair. Changing variables from (u, v) to the (θ, z) of Eq. (8), $\theta = (u \cos \phi - v \sin \phi) / r$ and $z = u \sin \phi + v \cos \phi$, the chain rule gives $\frac{\partial}{\partial u} = \frac{\cos \phi}{r} \frac{\partial}{\partial \theta} + \sin \phi \frac{\partial}{\partial z}$ and $\frac{\partial}{\partial v} = -\frac{\sin \phi}{r} \frac{\partial}{\partial \theta} + \cos \phi \frac{\partial}{\partial z}$, and combining them in the two orthogonal ways collapses each to a single term:

$$\boxed{\vec{n} \cdot \nabla = \cos \phi \frac{\partial}{\partial u} - \sin \phi \frac{\partial}{\partial v} = \frac{1}{r} \frac{\partial}{\partial \theta}, \quad \vec{m} \cdot \nabla = \sin \phi \frac{\partial}{\partial u} + \cos \phi \frac{\partial}{\partial v} = \frac{\partial}{\partial z},} \quad (126)$$

with $\nabla := (\frac{\partial}{\partial u}, \frac{\partial}{\partial v})$ throughout. The cross terms cancel by $\sin \phi \cos \phi - \cos \phi \sin \phi = 0$ in the first and add to $\sin^2 \phi + \cos^2 \phi = 1$ in the second. Note the direction of the substitution: these follow from expressing θ and z in terms of u and v , not from reading $\sin \phi$ and $\cos \phi$ off Eq. (6). Geometrically, $\vec{m}$ is the direction along which θ is stationary — a step $\varepsilon \vec{m}$ gives $d\theta = 0$ and $dz = \varepsilon$ — which is the surface's straight ruling of Section 2.3.

What the balance delivers. Under assumptions 2 and 3 the $\Delta \kappa_{ij}$ depend on x alone, so the only (u, v) dependence in Eq. (120) is through $\Delta \sigma_{rr}$, and because M_{uv} carries no pressure term the moment divergence collapses to a gradient:

$$\nabla \cdot \mathbf{M} := \begin{pmatrix} \frac{\partial M_{uu}}{\partial u} + \frac{\partial M_{uv}}{\partial v} \\ \frac{\partial M_{uv}}{\partial u} + \frac{\partial M_{vv}}{\partial v} \end{pmatrix} = \frac{\nu t^2}{12(1-\nu)} \nabla \Delta \sigma_{rr}. \quad (127)$$

Eq. (122) is then $\nabla \cdot \mathbf{M} + \mathbf{K} \vec{\Xi} = \vec{Q}$, and projecting it on the two basis directions of Eq. (125) splits it cleanly. The axial projection loses the curvature term entirely, because $\vec{m}^\top \mathbf{K} = \vec{0}$, and using

Eqs. (127)–(126),

$$\boxed{Q_z = \vec{m} \cdot (\nabla \cdot M) = \frac{\nu t^2}{12(1-\nu)} \frac{\partial \Delta \sigma_{rr}}{\partial z}, \quad Q_\theta = \vec{n} \cdot (\nabla \cdot M) + \frac{\Xi_\theta}{r},} \quad (128)$$

where Q_θ, Q_z and Ξ_θ are the projections $\vec{n} \cdot \vec{Q}$, $\vec{m} \cdot \vec{Q}$, $\vec{n} \cdot \vec{\Xi}$. The first is the classical plate relation between a moment gradient and a transverse shear force, recovered here with no curvature correction; the second relates the circumferential shear force to the circumferential first moment and does not determine either on its own.

So Eq. (128) closes the two-equations-in-four-unknowns problem only halfway: it *determines* Q_z outright, and it ties Q_θ to Ξ_θ without fixing either. That is the whole yield of the moment balance, and it is why the ledger shows no change across this row. It is worth being explicit about the trap here, because $\det K = 0$ invites the opposite conclusion. A singular system $K\vec{\Xi} = \vec{b}$ would indeed require $\vec{b}$ to lie in the range of K — the circumferential direction — and its axial projection would then be a condition on $\vec{b}$, hence on $\frac{\partial \Delta \sigma_{rr}}{\partial z}$. But Eq. (122) is not that system: the right-hand side is $\vec{Q} - \nabla \cdot M$, and $\vec{Q}$ is itself unknown. The axial projection is therefore an equation *for* Q_z , not a condition on the pressure, and it is only the Q_i that Section 3.4 discarded which makes the difference. The conditions that do constrain the problem come from the full thickness profile, which is the next subsection.

11.7 The in-plane equations, and the transverse shear profiles

This is where the even- ζ half of the $\hat{u}$ and $\hat{v}$ equilibrium equations is spent. Section 11.6 took the first moment and found it determines the shear forces; taking the equations pointwise instead gives the two conditions that actually constrain the problem, and the shear profiles along with them.

The interface condition, not equilibrium, sets the face values. Coulomb’s law does not set them: it bounds them,

$$\boxed{\sqrt{\sigma_{ur}^2 + \sigma_{vr}^2} \leq \mu |\sigma_{rr}| \quad \text{on each contact face,}} \quad (129)$$

with equality only where the interface actually slips. A friction-free interface is the statement that both faces sit at the lower end of that bound,

$$\boxed{\sigma_{ur}(\pm \tfrac{t}{2}) = \sigma_{vr}(\pm \tfrac{t}{2}) = 0,} \quad (130)$$

which is an admissible state of the frictional problem for any $\mu > 0$: the inequality (129) is satisfied *strictly*, so this is not an idealization outside the model but the particular equilibrium in which every interface sits inside its cone. It is also the exact condition at the two ends of the recursion, where Section 5.4 shows there is no neighbour to transmit anything. Note that Eq. (130) is four conditions, not two.

The pointwise equations are a two-point problem in ζ . Keeping every u and v derivative, Eqs. (20)–(21) are a first-order system across the thickness for the pair $\vec{\tau} := (\sigma_{ur}, \sigma_{vr})^\top$, driven by the in-plane stress gradients:

$$\boxed{\frac{\partial \vec{\tau}}{\partial \zeta} + \mathbf{K} \vec{\tau} = -\vec{s}(\zeta), \quad \vec{s} := \begin{pmatrix} \frac{\partial \sigma_{uu}}{\partial u} + \frac{\partial \sigma_{uv}}{\partial v} \\ \frac{\partial \sigma_{uv}}{\partial u} + \frac{\partial \sigma_{vv}}{\partial v} \end{pmatrix} = \bar{\vec{s}} + \zeta \vec{s}',} \quad (131)$$

with $\mathbf{K}$ from Eq. (123). The source is exactly affine in ζ because the in-plane stresses are, and its slope is the moment divergence: since $\sigma_{ij} = \bar{\sigma}_{ij} + \zeta \frac{12M_{ij}}{t^3}$ by Eq. (118),

$$\vec{s}' = \frac{12}{t^3} \nabla \cdot \mathbf{M}. \quad (132)$$

Projecting Eq. (131) on $\vec{n}$ and $\vec{m}$ decouples it *exactly*, because $\mathbf{K}\vec{\tau} = \vec{n}\sigma_{\theta r}/r$ by Eq. (125). Writing $\sigma_{\theta r} = \vec{n} \cdot \vec{\tau}$, $\sigma_{zr} = \vec{m} \cdot \vec{\tau}$, $s_\theta = \vec{n} \cdot \vec{s}$ and $s_z = \vec{m} \cdot \vec{s}$,

$$\boxed{\frac{\partial \sigma_{\theta r}}{\partial \zeta} + \frac{\sigma_{\theta r}}{r} = -s_\theta(\zeta), \quad \frac{\partial \sigma_{zr}}{\partial \zeta} = -s_z(\zeta).} \quad (133)$$

Each is first order with a prescribed value at *both* faces, so each carries one solvability condition and then determines its own profile. It is essential not to drop the source here: doing so would make both right-hand sides vanish and the profiles with them, and that conclusion would be false.

The solvability conditions are the in-plane equilibrium equations. For the axial component, differencing the two face values integrates the second of Eqs. (133) directly:

$$0 = \sigma_{zr}\left(\frac{t}{2}\right) - \sigma_{zr}\left(-\frac{t}{2}\right) = - \int_{-t/2}^{t/2} s_z d\zeta = -t \bar{s}_z \quad \implies \quad \bar{s}_z = 0, \quad (134)$$

exactly, with no correction. For the circumferential component the integrating factor is $e^{\zeta/r}$; requiring the solution launched from $\zeta = -t/2$ to arrive at zero gives $\int_{-t/2}^{t/2} e^{\zeta/r} s_\theta d\zeta = 0$, and expanding the exponential,

$$\bar{s}_\theta = -\frac{t^2}{12r} s'_\theta \left(1 - \frac{t^2}{60r^2}\right) = -\frac{t^2}{12r} s'_\theta + O\left(\frac{t^4}{r^3}\right). \quad (135)$$

These two are the answer to why the in-plane equations do not reappear as constraints later: they appear *here*, and what they say is that the thickness-averaged in-plane equilibrium is source-free to within a term of order t^2/r . The residual is available in closed form. Combining Eqs. (132), (127) and (126),

$$\bar{s}_z = 0, \quad \bar{s}_\theta = -\frac{\nu t}{12(1-\nu)r^2} \frac{\partial \Delta \sigma_{rr}}{\partial \theta}, \quad (136)$$

a small in-plane body force, purely circumferential, generated by the circumferential pressure gradient. Against the friction body force of Eq. (89), of order $\mu \Delta \sigma_{rr}/t$, it is smaller by $O(t^2/r\ell)$ with ℓ the length over which $\Delta \sigma_{rr}$ varies — one turn at most. It is discarded by assumption 9, the same

assumption that discarded the curvature weights of Eq. (115), leaving

$$\boxed{\frac{\partial \bar{\sigma}_{uu}}{\partial u} + \frac{\partial \bar{\sigma}_{uv}}{\partial v} = 0, \quad \frac{\partial \bar{\sigma}_{uv}}{\partial u} + \frac{\partial \bar{\sigma}_{vv}}{\partial v} = 0.} \quad (137)$$

The profiles, which are not zero. With the solvability conditions imposed, Eqs. (133) determine the profiles, and they are worth recording because they are the reason assumption 5 must be kept away from these two components. Integrating the axial equation with $\bar{s}_z = 0$,

$$\boxed{\sigma_{zr}(\zeta) = \frac{s'_z}{2} \left(\frac{t^2}{4} - \zeta^2 \right), \quad \bar{\sigma}_{zr} = \frac{t^2}{12} s'_z, \quad Q_z = \frac{t^3}{12} s'_z,} \quad (138)$$

a parabola vanishing at both faces with a *nonzero* mean — the classical Kirchhoff transverse shear, which no linear profile can represent. Substituting Eq. (132) gives $Q_z = \vec{m} \cdot (\nabla \cdot M)$, in exact agreement with Eq. (128): the moment balance and the pointwise profile are two readings of the same equation, and their agreement is a check on both. The circumferential profile is the same parabola to leading order, skewed by $O(t/r)$ through the exponential factor and carrying the small offset that Eq. (135) requires.

So at this stage σ_{ur} and σ_{vr} are *not* zero, and neither is the vector $\vec{\tau}$. Wherever the in-plane field has a gradient the shears carry a parabolic profile with a nonzero resultant, exactly as a transversely loaded plate does; Eq. (130) pins their boundary values and Eqs. (134)–(135) constrain the source, but nothing so far removes them. Four unknowns remain four. The profiles collapse only when the source vanishes *entirely*, slope as well as average, and establishing that is the business of Section 11.8.

Why this section must be friction-free. The choice was forced, and it is Eq. (137) that forces it. A sliding interface carries $\sigma_{ur}(\pm t/2) = -\mu \sigma_{rr}^\pm f_u$ from Eqs. (47)–(48), which is exactly the inhomogeneous face data Eq. (130) excludes; the boundary term of Eq. (28) then survives, and carried through Eq. (131) the interface shear becomes the distributed in-plane body force of Eq. (89), so that Eq. (137) acquires a right-hand side. Everything the next subsection concludes from source-freeness then fails. Friction is precisely a mechanism for making the in-plane problem inhomogeneous, which is why it requires the two-dimensional treatment of Section 8 and cannot be had here.

11.8 Saint-Venant: why the in-plane field is uniform

Assumption 10 asserts that u and v derivatives may be dropped. For the in-plane stresses that is not an independent assertion but a consequence of Eq. (137), and this subsection is the argument.

Collect what the in-plane problem now consists of. Two source-free equilibrium equations (137); one compatibility equation (34), which is available because Section 4.2 showed its curvature source vanishes with K ; and boundary conditions — traction-free long edges $\bar{\sigma}_{vv} = \bar{\sigma}_{uv} = 0$ at $v = \pm W/2$, and whatever the grips impose at $u = 0, L$. That is precisely the classical problem of a long rectangular strip in plane stress with no body force.

The uniform uniaxial field

$$\boxed{\bar{\sigma}_{uu} = \frac{N}{Wt} = \text{const}, \quad \bar{\sigma}_{vv} = \bar{\sigma}_{uv} = 0,} \quad (139)$$

with N the axial force resultant carried along the ribbon, satisfies every one of them *exactly*: equilibrium because all its derivatives vanish, compatibility because Eq. (34) contains only second derivatives, the long-edge conditions because two of its three components are identically zero, and the end conditions in the sense of resultants. It is the Saint-Venant solution of the strip.

Elastostatics is unique, so any other solution differs from Eq. (139) by a field with *zero* traction resultant at each end — a self-equilibrated end load. Saint-Venant’s principle, in the form Toupin made precise, is the statement that such a field decays exponentially into the interior, and for a traction-free strip the rate is known exactly. The decaying solutions of Eqs. (137) and (34) on $|v| \leq W/2$ are the Papkovitch–Fadle eigenfunctions $e^{-2\Lambda u/W}$, whose exponents satisfy $\sin 2\Lambda + 2\Lambda = 0$ for the modes symmetric across the width, and the slowest of these is

$$\boxed{\Lambda_1 = 2.1062 + 1.1254i \quad \Rightarrow \quad \text{amplitude} \sim e^{-4.212u/W},} \quad (140)$$

a decay length of $0.237W$. The end disturbance is therefore below one percent of its edge value within $1.09W$ of either grip.

Three conclusions follow, and they are the substance of assumption 10.

- Over the interior, $\bar{\sigma}_{uu}$ is constant and $\bar{\sigma}_{vv}$ and $\bar{\sigma}_{uv}$ vanish, so every $\frac{\partial}{\partial u}$ and $\frac{\partial}{\partial v}$ of an in-plane stress is zero. Nothing was assumed away; the equations were solved.
- The source $\vec{s}$ of Eq. (131) vanishes *entirely*, not just in its average. Its average was already zero by Eqs. (134)–(135); its slope is $\frac{12}{t^3}\nabla \cdot M$ by Eq. (132), which by Eq. (127) is proportional to $\nabla \Delta \sigma_{rr}$, and $\Delta \sigma_{rr} = t \sigma_{\theta\theta}/r$ from Eq. (142) below is uniform once $\sigma_{\theta\theta}$ is. So $\vec{s}' = 0$ as well, and Eq. (133) becomes homogeneous. An exponential and a constant cannot vanish at one ζ without vanishing at every ζ , so Eq. (130) now gives

$$\boxed{\sigma_{ur} = \sigma_{vr} = 0 \quad \text{throughout the thickness,}} \quad (141)$$

and with them $Q_\theta = Q_z = \Xi_{ur} = \Xi_{vr} = 0$. Four unknowns are gone — but only here, and only in the interior. This is the step Section 11.7 could not take on its own.

- The interior extent over which this holds is L less two end zones of order W , a fraction $1 - O(W/L)$ of the ribbon, and $W \ll L$ is the first item of Section 1.1. The approximation is charged to the geometry, not to a numerical accident.

Flagged step. The end zones are not negligible for the coil of Section 9: at the one-percent level they occupy $2 \times 1.09W/L = 8.7\%$ of the length. Two things limit the damage. The self-equilibrated correction enters the energy quadratically, since its first-order cross term against Eq. (139) involves its vanishing resultant, so an $O(1)$ end disturbance over 9% of the length costs much less than 9% of

U . And the measured force is a *resultant*, Eq. (90), which Saint-Venant's principle leaves exactly unchanged: the end zone redistributes $\bar{\sigma}_{uu}$ across the width without altering $\int \bar{\sigma}_{uu} dv$. **Downstream consequence:** $F(x)$ is unaffected at leading order, while the amplitude of Eq. (153) carries an end-zone uncertainty of the same few percent as the free-edge boundary layer of Section 11.11. Pointwise stress predictions near either grip are outside the closed form altogether, and belong to Section 9.

11.9 Equilibrium: the last reduction, and the step that fails

Of the three equilibrium equations, the $\hat{u}$ and $\hat{v}$ pair has now been spent twice over — its odd- ζ half in Section 11.6 and its even- ζ half in Sections 11.7–11.8 — so only the radial equation (22) remains unused, and it is the equation for the two pressure coefficients. With the shears gone by Eq. (141) and the in-plane field uniform by Eq. (139) it reads

$$\boxed{\frac{\partial \sigma_{rr}}{\partial \zeta} = \kappa_{uu} \sigma_{uu} - 2\kappa_{uv} \sigma_{uv} + \kappa_{vv} \sigma_{vv} = \frac{\sigma_{\theta\theta}}{r}}, \quad (142)$$

the middle expression being the membrane stress resolved onto $\hat{\theta}$. A layer in hoop tension presses inward on whatever lies beneath it, and integrating across one thickness gives the jump $\Delta \sigma_{rr} = t \sigma_{\theta\theta}/r$ — which is Eq. (60) again, now with its transverse-shear terms justified rather than assumed away. That determines $\Delta \sigma_{rr}$ from quantities already known.

The first ζ -moment of the same equation determines the quadratic coefficient. Multiplying Eq. (22) by ζ and integrating, the by-parts term is $\int \zeta \frac{\partial \sigma_{rr}}{\partial \zeta} d\zeta = \frac{t}{2} [\sigma_{rr}^+ + \sigma_{rr}^-] - t \bar{\sigma}_{rr} + \frac{P t^3}{12}$, whose first two pieces cancel by the definition of $\bar{\sigma}_{rr}$ as the surface mean, leaving

$$\boxed{\frac{P t^3}{12} = \kappa_{uu} M_{uu} - 2\kappa_{uv} M_{uv} + \kappa_{vv} M_{vv} - \frac{\partial \Xi_{ur}}{\partial u} - \frac{\partial \Xi_{vr}}{\partial v}}. \quad (143)$$

So P is fixed by the hoop projection of the bending moments, and it does not vanish: the moments of Eq. (120) are nonzero, so the coil carries a pressure bump through its own thickness even when both contact faces are unloaded. Section 11.14 evaluates it and finds it worth 1.1×10^{-4} of F . One unknown left: $\bar{\sigma}_{rr}$, the level about which the jump sits, and it needs a boundary value.

Flagged step. This is where the chain does not close. The contact recursion has *two* free ends, not one: Section 5.4 identified the base case at the outer face, where no material is wound beyond the last turn, but the innermost turn's inner face is equally free — there is a hole. Integrating Eq. (142) across the whole stack between two zero-pressure faces gives a solvability condition,

$$\int_{R_{\text{in}}}^{R_{\text{out}}} \sigma_{\theta\theta} \frac{dR}{R} = 0, \quad (144)$$

which a uniform, one-signed $\sigma_{\theta\theta}$ cannot satisfy. Under assumption 10 there is no room for the pressure to accumulate turn by turn at all, so the two conditions collapse onto the same point and over-determine $\bar{\sigma}_{rr}$ by one. The physical content is not obscure: a self-supporting stack of tensioned

layers with nothing inside it would collapse, so the hoop stress must change sign somewhere across the build, with the outer turns pressing inward and the inner ones pressing outward. That is the classical wound-roll distribution, and representing it requires the radius to vary across the stack, which assumption 3 forbids.

Which equation is given up. It is worth being exact about this, because the recursion (53) is the one relation in the whole chain that carries a (u, v) derivative, and assumption 10 removes derivatives. The closed form takes $\bar{\sigma}_{rr} = \Delta\sigma_{rr} = 0$, so that σ_{rr} is the quadratic bump alone. That choice satisfies the recursion *identically* — both sides are zero, since $\Delta\sigma_{rr} = 2\pi \frac{\partial \bar{\sigma}_{rr}}{\partial \theta}$ and there is no θ gradient to take; it satisfies *both* free-end conditions of the chain identically, because $\zeta^2 - t^2/4$ vanishes at both faces; and it satisfies the first ζ -moment (143) identically, which is what makes P nonzero. Exactly one scalar equation is violated: the *zeroth* ζ -moment of Eq. (142), which demands $\Delta\sigma_{rr} = t \bar{\sigma}_{\theta\theta}/r$ and gets 0. The residual is $\bar{\sigma}_{\theta\theta}/r$.

Downstream consequence. $\bar{\sigma}_{rr}$ enters the energy only through the strictly positive quadratic term of Eq. (114) and, via Eq. (112), through the Poisson relief of the axial stress. Both effects are therefore *bounded* by the largest admissible $|\bar{\sigma}_{rr}|$, which is what makes the failure survivable. Section 11.13 evaluates two closures — $\bar{\sigma}_{rr} = 0$ as above, and the full accumulation imported from the recursion — and finds the energy to differ by at most 8×10^{-6} and F by at most 1.5×10^{-5} . Everything from Eq. (153) onward is insensitive to the choice. The one place it is *not* innocuous is friction, because the frictional traction is proportional to the surface pressure that this relaxation sets to zero; Section 11.16 takes that up.

How the missing condition could be supplied. Three routes exist, and pricing them is the honest way to bound what has been lost. Each is measured against the membrane energy it competes with.

A hub. If the innermost turn bears on a rigid cylinder of radius $r(x)$, then the reaction at $u = 0$ carries a distributed pressure alongside the axial $-F$ that prevents rigid-body motion, and Eq. (144) becomes a determination of it,

$$\sigma_{rr}^-|_{\text{inner}} = - \int_{R_{\text{in}}}^{R_{\text{out}}} \sigma_{\theta\theta} \frac{dR}{R} = -\Pi \bar{\sigma}_{uu}, \quad \Pi := \frac{t n}{r} \cos^2 \phi = \frac{2 \rho_{\text{jam}} \rho}{\lambda^2}. \quad (145)$$

The reaction is determined rather than free, so no constant enters; it is compressive, so it satisfies assumption 7 identically; and because the hub radius tracks $r(x)$ rather than obstructing it, the hub does no work as the coil tightens, so Eq. (67) is untouched. What it costs is physical: Eq. (145) is an axisymmetric pressure, needed at every θ , and the clamp at $u = 0$ acts at one θ and cannot supply it. This closure therefore describes a ribbon wound on a hub, which is a different experiment from one wound with a hollow core, and directly testable.

Redistributing the axial strain. A free-standing coil can instead let the hoop stress change sign, which is what Eq. (144) asks for. The axial strain must then satisfy two integral conditions at once, $\langle \bar{\epsilon}_{uu} \rangle = \epsilon$ from the arc length and $\langle \bar{\epsilon}_{uu} R^{-2} \rangle = 0$ from Eq. (144), and the minimum-energy

field doing so is $\bar{\epsilon}_{uu} = \alpha + \beta/R^2$, whose energy exceeds the uniform value by

$$\mathcal{K} = \left[1 - \frac{g \ln^2 g}{(g-1)^2} \right]^{-1} \approx \frac{3\rho^2}{4\rho_{\text{jam}}^2}, \quad g := \left(\frac{R_{\text{out}}}{R_{\text{in}}} \right)^2. \quad (146)$$

The penalty is severe precisely because the two weights, 1 and R^{-2} , differ by only $g-1 \approx 4\rho_{\text{jam}}/\rho$ across a thin build: nearly parallel constraints are expensive to satisfy together. For the coil of Section 9 $\mathcal{K}$ runs from 455 at small x to 12 at $x/L = 0.99$, the closed form on the right reproducing the exact expression to better than one percent. The profile does change sign, at a neutral radius just inside the mean, with an amplitude of ± 7 to ± 38 times ϵ . The picture is right; the price is not, since this route alone would raise F by 55% at $x/L = 0.99$.

A width-wise shear. A cross-section transmits σ_{uv} as well as σ_{uu} , and σ_{uv} may carry a profile across the width with a nonzero resultant while still vanishing at the free edges $v = \pm W/2$, exactly as the transverse shear in a beam does. Since $\sigma_{\theta\theta}$ contains $-2\kappa_{uv}\sigma_{uv}$, a shear of

$$\bar{\sigma}_{uv} \approx \frac{1}{2} E \epsilon \cot \phi = \frac{E \epsilon \rho}{2\chi} \quad (147)$$

nulls the mean hoop stress and satisfies Eq. (144) outright, at a cost of $\frac{(1+\nu)\cot^2\phi}{2}$ relative to the membrane term — 15.6 at $x/L = 0.2$, falling to 0.017 at $x/L = 0.99$. That is two orders of magnitude cheaper than $\mathcal{K}$ everywhere, so it is the route the coil takes, and it is why the omission is survivable: adding Eq. (147) to the energy moves the minimizing radius by at most 0.16% and F by at most 0.42%. The field is two-dimensional in (u, v) and so lies outside assumption 10, but inside what Section 9 carries.

Two corrections to the natural reading. Bending is not what pushes the inner turns outward: for a circular ring the moment is uniform in θ , so $\frac{\partial M}{\partial s} = 0$, the transverse shear vanishes and radial equilibrium reduces to $\kappa N = p$. A uniform bending moment is self-equilibrated and exerts no distributed radial load; what pushes outward is hoop *compression*. Bending decides which radius the coil prefers, and so — through the arc length — what N is, but it does not push. And the packing scale does not rescue Eq. (144): ρ_{jam} enters only through Π and $\mathcal{K}$, altering magnitudes without touching the sign structure.

Flagged step. There is reason to suspect Eq. (144) of overstating the difficulty. In the membrane-only limit, stationarity of the energy in ρ and the integrated radial equilibrium are not merely consistent but *identical*: both reduce to $\epsilon = 0$, i.e. $\rho = \sqrt{1 - \chi^2}$, and the same term-by-term identity $\frac{\partial U_{\text{mem}}}{\partial r} \propto \sum_k \sigma_{\theta\theta,k}/R_k$ holds even when a radius distribution is carried across the stack. The two conditions diverge only once bending is included, and they diverge by exactly the bending term — the one term a linearization about a *fixed* geometry cannot see, since Eqs. (20)–(22) are written at fixed κ and the energy cost of changing κ is a geometric-stiffness effect of the next order. The apparent over-determination may therefore be a missing term in Section 3 rather than a gap in the physics. This is not settled here. **Downstream consequence:** none quantitative, because all three closures above bracket F within half a percent, but it is the one place where the equilibrium and energy formulations of this model are known to disagree, and the natural next thing to resolve.

11.10 Compatibility, and what is left of it

In the friction subsystem, compatibility is the equation that closes the in-plane problem (Section 8.1). Here it has already been spent twice, and it is worth accounting for both.

The first is easy to miss. Eq. (107) is not a constitutive or an equilibrium statement but a compatibility statement: it is what normality plus the existence of a mid-surface displacement field forces the through-thickness variation to be. The second is Section 11.8, which used Eq. (34) as one of the three governing equations of the strip — without it the uniform field (139) would not be the unique interior solution, and the Papkovitch–Fadle spectrum that sets the decay rate would not exist. That the uniform field then satisfies Eq. (33) identically, every term being a second derivative of a constant, is the *consequence* of having solved it rather than a reason to have ignored it.

What remains is the part of three-dimensional compatibility that the in-plane relation never contained, because ϵ_{rr} was not part of Section 4.2. Let w be the displacement along $\hat{r}$, so $\epsilon_{rr} = \frac{\partial w}{\partial \zeta}$; then Eq. (105) is itself two of the six three-dimensional Saint-Venant conditions, and reads

$$\frac{\partial a}{\partial \zeta} = -\frac{\partial w}{\partial u} + \kappa_{uu}a - \kappa_{uv}b, \quad \frac{\partial b}{\partial \zeta} = -\frac{\partial w}{\partial v} - \kappa_{uv}a + \kappa_{vv}b, \quad (148)$$

with a, b the in-plane displacements of Section 4.3. The content is that a through-thickness strain varying over the sheet tilts the normal, and a tilted normal is a curvature change: $\frac{\partial^2 w}{\partial u^2}$ adds to $\Delta\kappa_{uu}$, so a pressure gradient bends the ribbon. Under assumption 10 every derivative on the right vanishes, the remaining conditions reduce to identities, and ϵ_{rr} is left free to vary with ζ alone — which is exactly what Eq. (113) asks of it, since σ_{rr} is affine in ζ .

So the residual three-dimensional compatibility conditions are consistent but idle: they constrain nothing the interior solution has not already settled. With equilibrium and compatibility both exhausted, the remaining closure must come from the boundary conditions, and it does.

11.11 The boundary conditions close the system

Three unknowns remain: $\bar{\epsilon}_{uu}$, $\bar{\epsilon}_{vv}$, $\bar{\epsilon}_{uv}$, now constants rather than fields. Three conditions remain unused.

The long edges are traction-free, which for a plate means the vanishing of the stress resultants,

$$\bar{\sigma}_{vv} = \frac{1}{t} \int_{-t/2}^{t/2} \sigma_{vv} d\zeta = 0, \quad \bar{\sigma}_{uv} = \frac{1}{t} \int_{-t/2}^{t/2} \sigma_{uv} d\zeta = 0. \quad (149)$$

These must be read as resultants and not pointwise: imposing $\sigma_{vv} = 0$ at every ζ would require $\Delta\kappa_{vv}$ to adjust, and $\Delta\kappa_{vv}$ is fixed by the winding, so the edge moment M_{vv} of Eq. (120) is a reaction to assumptions 2 and 3 rather than something the sheet is free to relax. **A real ribbon does relax it, in a boundary layer of width $\sim \sqrt{rt}$ along each long edge — at most 0.14 cm against a width of 8 cm for the coil of Section 9. Downstream consequence: the edge strip contributes an $O(\sqrt{rt}/W)$**

Table 1: The elimination ledger. “In energy” counts only the fields appearing in Eq. (104) after the workless pair is removed.

step	mechanism	unknowns	in energy
Eq. (104)	—	12	12
Eq. (105)	assumption 8, normality	10	8
Eqs. (107), (108)	assumption 5, thickness ansatz	9	5
Eqs. (112)–(113)	constitutive law	9	5
Eq. (122)	first moment: no reduction, supplies $\nabla \cdot M$	9	5
Eqs. (134)–(135)	in-plane equilibrium, free faces	9	5
Eqs. (139), (141)	Saint-Venant interior $\Rightarrow \vec{\tau} \equiv 0$	5	5
Eq. (142)	radial equilibrium	4	4
Eq. (144)	over-determined; bounded	3	3
Eq. (149)	free long edges	1	1
Eq. (152)	arc length	0	0

fraction of the width at a different energy density, so Eq. (153) is accurate to a couple of percent in amplitude, not better. This is the largest single uncertainty in the closed form.

Because the bending parts of Eq. (107) and the pressure gradient are both odd in ζ , Eq. (118) removes them from both resultants and Eq. (149) becomes two algebraic relations among mid-surface quantities alone. Solving them,

$$\boxed{\bar{\epsilon}_{vv} = -\nu \bar{\epsilon}_{uu} - \frac{\nu(1+\nu)}{E} \bar{\sigma}_{rr}, \quad \bar{\epsilon}_{uv} = 0.} \quad (150)$$

The first term is the ordinary Poisson contraction under axial tension; the second is its pressure counterpart, with the sign one expects — with $\bar{\sigma}_{rr} \leq 0$ the ribbon is squeezed through its thickness and spreads in width. Substituting Eq. (150) back into Eq. (112) gives the reduced constitutive relation this section will use,

$$\boxed{\bar{\sigma}_{uu} = E \bar{\epsilon}_{uu} + \nu \bar{\sigma}_{rr},} \quad (151)$$

exact rather than approximate. Because $\bar{\sigma}_{rr} \leq 0$, the coil’s own contact pressure *lowers* the tension needed to hold a given elongation: squeezing the ribbon radially makes it longer by Poisson expansion, which supplies part of the arc length for free. That is the soda-can effect made quantitative, and Eq. (145) sizes it at $\nu \Pi/2 \leq 1.2\%$.

The third condition is the arc-length constraint (2), which applies to the mid-surface material line and so fixes

$$\boxed{\bar{\epsilon}_{uu} = \lambda - 1 = \sqrt{\rho^2 + \chi^2} - 1,} \quad (152)$$

the same normalization already imposed in Eq. (93). Three conditions for three unknowns: the ledger closes.

11.12 The closed-form energy, and the derivative that gives F

Every field is now an expression in ρ and χ . Substituting Eqs. (150) and (152) into Eq. (114), integrating over ζ with Eq. (118), and using the two identities of Eq. (55) to collapse the curvature invariants,

$$\boxed{\frac{U}{E L W t} = \frac{\tau^2}{24(1-\nu^2)} \left[\underbrace{\left(\frac{1}{\rho} - 1\right)^2}_{\text{bending, hoop}} + \underbrace{\frac{2(1-\nu)\chi^2}{\rho\lambda^2}}_{\text{bending, twist}} + \underbrace{\Gamma(\lambda-1)^2}_{\text{membrane}} \right], \quad \tau := \frac{t}{r_{\text{nat}}},} \quad (153)$$

which is exactly the two-mode energy $\tilde{U}$ of Eq. (65): the τ^2 prefactor reproduces the bending stiffness D and the group Γ term by term. Section 7.1 assumed this; it is now derived, and the derivation says what the assumption was worth. The two scalars of that section were not a lucky guess but the exact consequence of assumptions 2, 3, 5, 8, 9 and 10, and the only thing lost along the way was Eq. (116) plus the bounded pressure remainder.

Density or total? The question matters, and the answer is that the derivative must be taken of the total energy integrated over the *reference* volume. Two reasons. Eq. (114) is a linear-elastic strain energy density, whose integral is over the undeformed body by construction; and L, W, t are the reference dimensions listed in Section 1.1, hence constants, whereas the deformed dimensions depend on ρ and would contribute spurious terms. Having made that choice, the distinction becomes harmless: assumption 10 makes the density uniform, so $U = L W t u_{\text{el}}$ and differentiating the total or the density differs only by the constant factor $L W t$.

The geometry then follows from stationarity at fixed χ , which is Eq. (67) and its root $\rho(\chi)$ of Section 7.2, and the force from the envelope theorem of Section 7.4,

$$F(x) = \left. \frac{\partial U}{\partial x} \right|_{\rho} = E W t \left. \frac{\partial}{\partial \chi} \left(\frac{U}{E L W t} \right) \right|_{\rho}, \quad (154)$$

so that $\frac{d\rho}{dx}$ is never needed. Carrying Eq. (153) through Eq. (154) gives Eq. (97) with $\mathcal{A} = 1$, the friction-free limit of the measured force.

11.13 The energy-minimizing state

With the energy closed, every field at the energy-minimizing radius is a closed-form function of χ . Setting $\bar{\sigma}_{rr} = 0$, which is the closure assumption 10 permits, Eqs. (150)–(151) and (113) give the

complete state:

$$\begin{aligned}
& \bar{\epsilon}_{uu} = \epsilon, \quad \bar{\epsilon}_{vv} = -\nu \epsilon, \quad \bar{\epsilon}_{uv} = 0, \quad \bar{\epsilon}_{rr} = -\nu \epsilon, \\
& \bar{\sigma}_{uu} = E \epsilon, \quad \bar{\sigma}_{vv} = \bar{\sigma}_{uv} = 0, \quad \sigma_{ur} = \sigma_{vr} = \sigma_{rr} = 0, \\
& \sigma_{uu}(\zeta) = E \epsilon + \frac{E \zeta}{1 - \nu^2} (\Delta \kappa_{uu} + \nu \Delta \kappa_{vv}), \quad M_{ij} = D (\Delta \kappa_{ij} + \nu \Delta \kappa_{kk} \delta_{ij}) \text{ as in Eq. (120),}
\end{aligned} \tag{155}$$

with $\epsilon(\chi) = \lambda(\rho(\chi)) - 1$ from Section 7.2 and the $\Delta \kappa_{ij}$ from Eq. (54). Evaluated for the coil of Section 9, where $D = 0.0275 \text{ N cm}$:

x/L	ρ	ϵ	$\bar{\sigma}_{uu}$ [N cm ⁻²]	σ_{uu} bending at $\zeta = \frac{t}{2}$ [N cm ⁻²]	M_{uv} [N]	F [N]
0.40	0.917	1.2×10^{-6}	0.35	-25.6	0.0038	0.04
0.80	0.600	1.5×10^{-5}	4.57	-66.0	0.0077	0.34
0.947	0.322	2.0×10^{-4}	59	+130	0.0091	4.5
0.995	0.143	5.3×10^{-3}	1575	+984	0.0095	125

Three features are worth reading off. The membrane stress is dominated by the bending stress by a factor of 73 at $x/L = 0.4$, falling through 14 at 0.8 and 2.2 at 0.947 to 0.62 at 0.995: the ribbon's surface stress is a bending stress almost everywhere, and only at the very end of the pull does membrane tension take over. The bending stress changes sign near $x/L = 0.9$, where $\Delta \kappa_{uu} + \nu \Delta \kappa_{vv}$ passes through zero. And the strain stays below 10^{-4} until $x/L = 0.926$ and never reaches 10^{-2} , which is the *a posteriori* justification of the small-strain half of assumption 4 — the model's own output respects the linearity it assumed, up to the crossover of Section 7.3, past which it hands over to a large-strain regime it does not claim to describe.

Finally, the price of the two approximations this section made rather than derived. The discarded volume-element term of Eq. (116) is worth at most 1.0×10^{-3} of the energy, 2.1×10^{-4} of ρ and 1.6×10^{-3} of F . The unresolved pressure level of Section 11.9 is worth at most 8.3×10^{-6} of the energy and 1.5×10^{-5} of F under the recursion-informed closure, and 3.9×10^{-8} and 7.1×10^{-8} under the homogeneous one (Figure 8a). Both are far below the couple of percent that the free-edge boundary layer of Section 11.11 already concedes, which is therefore the honest accuracy of Eq. (153).

11.14 Results, and what the derivation was worth

Figure 9 collects the solution.

The minimum is unique, and for a structural reason. Panel (b) shows $U(\rho; \chi)$ at five frozen extensions, each with exactly one interior minimum; counting stationary points on a four-thousand-point grid gives one at every χ from 0.1 to 0.99. The reason is visible in Eq. (153): the hoop term $(1/\rho - 1)^2$ diverges as $\rho \rightarrow 0$, the membrane term $\Gamma(\lambda - 1)^2$ rises steeply as $\rho \rightarrow \sqrt{1 + \chi^2}$ where the ribbon runs out of slack, and both are convex in between. No second branch exists to be missed, which is worth knowing because the stationarity condition (67) is solved numerically.

The geometry barely leaves the inextensible curve. Panel (a): ρ_* sits almost on $\sqrt{1 - \chi^2}$ throughout, and never approaches $\rho_{\text{jam}} = 0.040$ — consistent with Section 7.3, where the packing scale was shown to be an asymptote reached only for windings with $\rho_{\text{jam}}/\rho_* > 1$, and this coil sits at 0.28. Panel (c): the strain crosses 10^{-4} at $x/L = 0.926$ and never reaches 10^{-2} , which is the *a posteriori* justification of the small-strain half of assumption 4 — the model’s own output respects the linearity it assumed.

The two force terms trade places, and the twist term turns over. Panel (d) decomposes F exactly as Eq. (97) does, with $\mathcal{A} = 1$. The twist term rises, saturates near 0.037 N and then *falls*: it carries the factor $\rho\chi/\lambda^4$, which peaks when $\rho \approx \chi$ and decays once the coil has tightened past that. The membrane term overtakes it at $x/L = 0.545$ and thereafter carries essentially all of the force. In the frictional problem the crossover sits earlier, near $x/L \approx 0.2$ (Section 8.4), because $\mathcal{A}$ multiplies the membrane term and not the twist term; the friction-free curve here is the lower envelope of that.

The through-thickness fields. Figure 10(a) shows that the surface stress is a *bending* stress almost everywhere: the membrane share of the peak runs from 0.01 at $x/L = 0.4$ to 0.54 at 0.99. Panel (b) shows the quadratic pressure of Eq. (143), which is present even though both contact faces carry zero pressure, and which changes sign near $x/L \approx 0.7$ where the hoop projection of the moment does. It is tensile at mid-thickness below that, which violates nothing: assumption 7 constrains the contact faces, not the interior of the material. Its cost is $|\Delta F/F| \leq 1.1 \times 10^{-4}$, so it belongs with the other bounded remainders of Section 11.13.

Where the fields are uniform, and a warning that is not about this section. Panel (c) maps the (u, v) plane. Inside the interior the stress field is uniform by construction — that is what Section 11.8 bought — so plotting a stress over it would show a single colour, and the panel maps what does vary instead.

Flagged, for Sections 8 and 9 rather than for this one. The contact gates of Eqs. (40)–(41) are far more restrictive at these parameters than Section 5.1 suggests. The axial offset between successive turns, $2\pi r \sin \phi$, peaks at 6.28 cm against a width of 8 cm, and the fraction of the ribbon with *both* neighbours present is 0.69 at $x/L = 0.1$, 0.10 at 0.3, and identically *zero* from $x/L \approx 0.45$ to 0.95; over that range the fraction with even one neighbour falls to 0.43. **Downstream consequence:** none for Eq. (153), which is friction-free and whose σ_{rr} is the bending bump rather than a contact pressure. But the friction subsystem has almost no contact area to work with over the middle of the pull at $W/r_{\text{nat}} = 4$, so the capstan gain of Eq. (96) and the accumulation exponent of Eq. (103) are being computed on a sparse contact set there. Widening the ribbon or reducing r_{nat} until $2\pi r_{\text{nat}} \ll W$ would remove the problem; the competing constraint of Section 9.3 is that $\mu L/r_{\text{nat}}$ must stay small for conditioning, so the two cannot both be relaxed freely.

11.15 Comparison with the two-mode model

It is worth stating plainly what this section changed about the prediction of Section 7: nothing. Eq. (153) is not close to Eq. (65), it is the same expression — the symbolic difference is zero, not small — and every number agrees to the last figure quoted: $\rho_* = 0.16206$ and $F = 75.257$ N at $x/L = 0.99$ from either route. A reader who stopped at Section 7.1 would have the right answer.

What the derivation bought is worth separating into three things.

The two scalars became a consequence. Section 7.1 wrote down a bending density and $\frac{1}{2}E\epsilon^2$ and proceeded. Here that form is the endpoint of a ledger (Table 1) that starts from $\frac{1}{2}\sigma_{ij}\epsilon_{ij}$ with twelve unknown fields, and every elimination is charged to a numbered assumption. The value of that is not reassurance but *traceability*: if a prediction disagrees with experiment, the ledger says which assumption to attack, and the two candidates it identifies — the free-edge boundary layer and the two-free-ends relaxation — are not the ones a reader would have guessed from Section 7.1.

An error budget appeared where there had been none. Ranked by size:

source	effect on F	where
free-edge boundary layer, width $\sqrt{rt}$	a few percent	Section 11.11
end zones, $2 \times 1.09 W/L = 8.7\%$ of the length	resultant-preserving, so ≈ 0	Section 11.8
two-free-ends relaxation (width-wise shear route)	4.2×10^{-3}	Section 11.9
discarded volume element $\frac{t^2}{12r}\bar{\epsilon}_{uu}\Delta\kappa_{uu}$	1.6×10^{-3}	Section 11.5
quadratic pressure P	1.1×10^{-4}	Section 11.14
contact pressure $\bar{\sigma}_{rr}$	1.5×10^{-5}	Section 11.9

The ordering is the useful part. Every term the three-dimensional treatment *restored* — the transverse stress state, the exact ζ integration, the contact recursion — lands at 10^{-4} or below, while the two largest entries are things no amount of care with the constitutive law would fix, because both are consequences of the uniform-field reduction rather than of the energy. The honest accuracy of Eq. (153) is therefore a couple of percent, set by the free-edge layer, and improving it means a two-dimensional solve and not a better energy.

Three errors upstream came to light. The derivation reached back into Sections 3–6 and corrected them. The first moment of in-plane equilibrium was missing the transverse shear force Q_i , because assumption 5 had been applied to σ_{ur} and σ_{vr} , where it forces $Q_i = 0$ outright (Section 3.4). A solvability condition $\frac{\partial\Delta\sigma_{rr}}{\partial z} = 0$ had been read off the singularity of $\mathbf{K}$, when with Q_i restored the same projection determines Q_z instead (Section 11.6). And σ_{zr} had been argued to vanish when in fact it is a parabola with a nonzero resultant wherever the in-plane field has a gradient (Section 11.7). All three sat upstream of Section 7, and Section 7 happened to depend on none of them — which is the reason its answer survives, and also the reason the errors went unnoticed.

The summary is that Section 7 was right for reasons it had not checked. This section checked them, found three faults elsewhere in the document, and produced the error bars the prediction had been missing. Had the answer moved, that would have been the more interesting outcome; it did not, and the identity of Eqs. (65) and (153) is a result in its own right.

11.16 Friction: why the closed form cannot be the measured force

Section 11.7 chose the lower corner of the Coulomb cone (129) and showed the choice was admissible. Admissible is not the same as actual, and it is worth being blunt about the difference, because it decides what Eq. (153) can and cannot predict.

Sliding means equality. A block of weight $\mathcal{W}$ resting on a surface carries a friction force anywhere in $[0, \mu\mathcal{W}]$, determined by whatever else is pushing it. A block *sliding* at constant velocity carries exactly $\mu\mathcal{W}$: the inequality is slack only while the interface is stuck. Our interfaces are not stuck. Section 5.2 shows the relative slip velocity (43) is nonzero for every $x > 0$, and assumption 6 adopts the kinetic law on exactly that ground. So in the physical, sliding coil

$$\boxed{\sqrt{\sigma_{ur}^2 + \sigma_{vr}^2} = \mu |\sigma_{rr}| \quad \text{on every contact face, not } \leq .} \quad (156)$$

The field of Section 11.7 is therefore one equilibrium of the frictional system — the one the coil would occupy if it could slide freely — and not the one it occupies while being pulled. Eq. (153) is the friction-free limit, and it is a *lower* bound on the measured force rather than an estimate of it.

Stored energy is no longer the whole balance. Friction dissipates, and dissipated work is not stored, so the envelope relation (154) is incomplete once Eq. (156) holds. The quasi-static work balance over an increment dx is

$$\boxed{F(x) dx = \underbrace{dU}_{\text{stored}} + \underbrace{d\mathcal{D}}_{\text{dissipated}} \implies F(x) = \frac{dU}{dx} + \mathcal{D}(x),} \quad (157)$$

with $\mathcal{D}$ the frictional dissipation per unit pull. No amount of care with the elastic energy will produce the second term; it has to be computed from the slip and the pressure.

There is an apparent conflict with Section 11.2 to clear up, since Eq. (106) said the transverse shears are workless. Both statements are true and they are about different work. Normality (105) is a statement about material lines within *one* layer, and it does make $\sigma_{ur}\epsilon_{ur} + \sigma_{vr}\epsilon_{vr}$ vanish pointwise, so friction contributes nothing to the stored energy density. The slip, however, is a displacement *discontinuity* across a contact interface, between two different layers, about which normality says nothing at all. So friction stores nothing and dissipates through a surface integral:

$$\boxed{\mathcal{D}(x) = \mu |\vec{v}_{\text{slip}}| \iint_{\text{contact}} |\sigma_{rr}| du dv, \quad |\vec{v}_{\text{slip}}| = \frac{2\pi}{\theta_L \cos \phi} = \frac{2\pi \lambda}{\theta_L \rho},} \quad (158)$$

the slip magnitude following from Eq. (44) in the inextensible limit and being, by Eq. (43), *independent of* (u, v) — every contact in the coil slides at the same rate, which is the one genuine simplification available here. It rises from $1/n$ at $x = 0$ to $2\pi\lambda/\theta_L\rho$ as the coil tightens, a factor of six over the pull.

How large is it? Large enough to dominate. Taking the pressure at the scale the recursion sets, $|\sigma_{rr}| \sim \frac{1}{2}\Pi \bar{\sigma}_{uu}$ with $\bar{\sigma}_{uu} = E\epsilon$ and *no* capstan amplification, and the contact area from the gates of Eqs. (40)–(41),

x/L	0.2	0.4	0.6	0.8	0.99
$ \vec{v}_{\text{slip}} $	0.064	0.069	0.079	0.105	0.389
contact fraction	1.00	0.85	0.49	0.49	1.00
$\mathcal{D}$ [N]	0.09	0.36	0.65	2.70	1140
$\frac{dU}{dx}$ [N]	0.016	0.039	0.090	0.329	75.3
$\mathcal{D}/\frac{dU}{dx}$	5.5	9.0	7.3	8.2	15

So the dissipative term exceeds the stored-energy term by between five and sixteen times *before* any amplification, and Section 8.4 reports a capstan gain of up to a hundred on top of that. The friction-free curve of Figure 9(d) is therefore not a modest underestimate of $F(x)$; it is an order of magnitude low over most of the pull. This is worth stating because the two curves have similar *shape* — both climb steeply near $x = L$ — and shape alone cannot be used to argue that friction is unimportant.

Why the pressure cannot simply be post-processed. It is tempting to take the converged $r(x)$ and $\epsilon(x)$ from Eq. (153), solve Eq. (142) and the recursion for $\sigma_{rr}(u, v; x)$, evaluate Eq. (158), and add the result to F . The geometry does survive that treatment — Section 11.9 bounds friction’s effect on ρ through the radial balance at $\mu \Pi \cos \phi/2\pi \lesssim 4 \times 10^{-3}$, so $r(x)$ and $\epsilon(x)$ may indeed be taken as given. The pressure does not. The reason is that σ_{rr} is not an input to friction but an output of it:

$$\sigma_{rr} \xrightarrow{\text{Eq. (156)}} \sigma_{ur}, \sigma_{vr} \xrightarrow{\text{Eq. (89)}} \bar{\sigma}_{uu} \xrightarrow{\text{Eq. (142)}} \Delta\sigma_{rr} \xrightarrow{\text{Eq. (53)}} \sigma_{rr}, \quad (159)$$

a closed loop with positive gain. Its growth rate is the accumulation exponent λ_+ of Eq. (103), and the loop is the entire content of Section 8. Solving the second link requires $\bar{\sigma}_{uu}(u, v)$, which is no longer uniform, so the whole subsystem is two-dimensional and cannot be reduced the way this section reduced the elastic problem.

What breaks in this section’s chain, item by item. Turning Eq. (156) on invalidates four specific steps above, and they are worth listing because each is a place where a frictional calculation must differ from this one.

1. Eq. (130) fails, so the boundary term of Eq. (28) no longer vanishes: the right-hand side of the first moment gains $-\frac{t}{2}\mu[\sigma_{rr}^+ + \sigma_{rr}^-]f_i = -\mu t \bar{\sigma}_{rr}f_i$ by Eq. (49).
2. The two-point problem (131) acquires inhomogeneous data, so the solvability conditions (134)–(135) no longer read “source-free”. The source becomes the friction body force (89), of order $\mu \Delta\sigma_{rr}/t$, which is larger than the $O(t^2/r)$ residual (136) by $12(1-\nu)\mu r \ell/\nu t^2$ — overwhelmingly.

3. With a distributed in-plane load, Section 11.8 fails at its first step: a strip carrying a body force has no uniform interior solution, so $\frac{\partial}{\partial u}$ and $\frac{\partial}{\partial v}$ cannot be dropped and assumption 10 is void.
4. $\Delta\sigma_{rr} \neq 0$ then makes the recursion (53) active rather than trivially satisfied, which restores the very (u, v) dependence assumption 10 removed — and with it the possibility of satisfying both free ends of the chain, since an accumulation that varies along θ can return to zero where a uniform one cannot.

Item 4 is the encouraging one: the obstruction of Section 11.9 is an artifact of the friction-free reduction, not of the physics, and the frictional problem does not inherit it.

An unused check. Eq. (157) and Eq. (90) are two independent routes to the same $F(x)$: one sums stored energy and dissipation over the whole coil, the other reads a resultant at $u = L$. Section 9 computes the second and not the first. Evaluating $\mathcal{D}$ from the converged pressure field and comparing $\frac{dU}{dx} + \mathcal{D}$ against $Wt [\bar{\sigma}_{uu}(L) \sin \phi + \bar{\sigma}_{uv}(L) \cos \phi]$ would be a global check on the friction subsystem of a kind it currently lacks, and would catch precisely the sign and conditioning failures reported in Section 9.3.

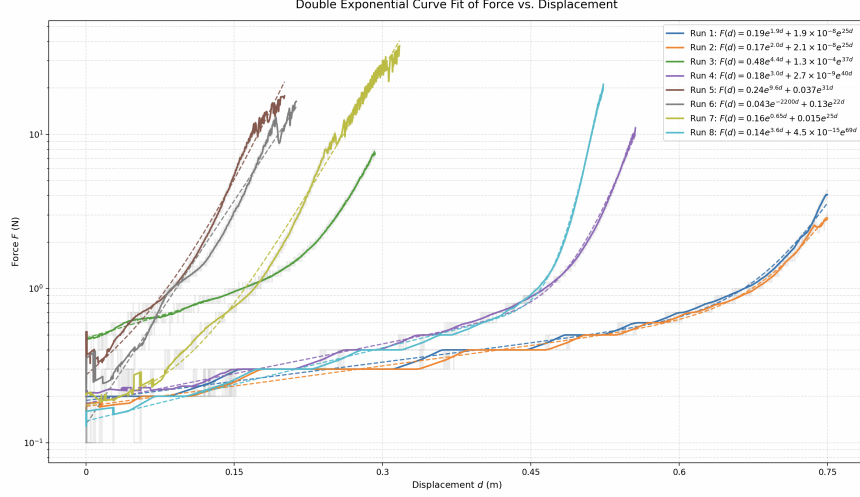


Figure 7: Instron force–displacement data: raw quantized data (grey), smoothed (solid), and curve fit (dashed).

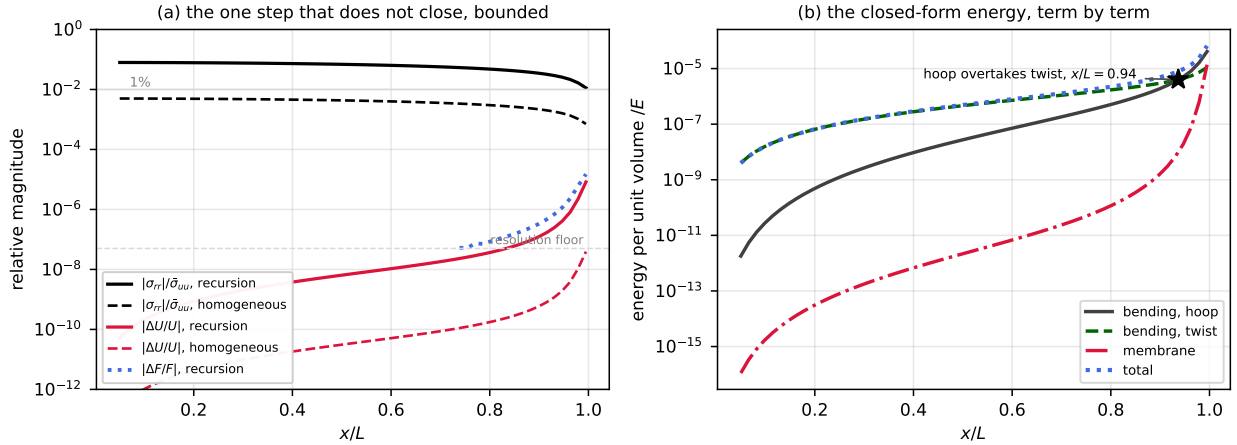


Figure 8: (a) The flagged step of Section 11.9, bounded. The transverse pressure is shown for the two closures — the single-layer jump that assumption 10 permits, and the full accumulation imported from the recursion — together with the change each makes to the energy and to F . The largest effect anywhere is 8×10^{-6} in the energy and 1.5×10^{-5} in the force, so the unresolved boundary value does not reach the deliverable. $|\Delta F/F|$ is drawn only above the sweep’s finite-difference floor. (b) The three terms of Eq. (153) evaluated along the minimizing path $\rho(\chi)$. Twist bending carries the energy for most of the pull, the hoop term overtakes it near $x/L = 0.94$, and the membrane term — negligible over four decades — climbs steeply enough at the end to reach a quarter of the bending energy by $x/L = 0.995$.

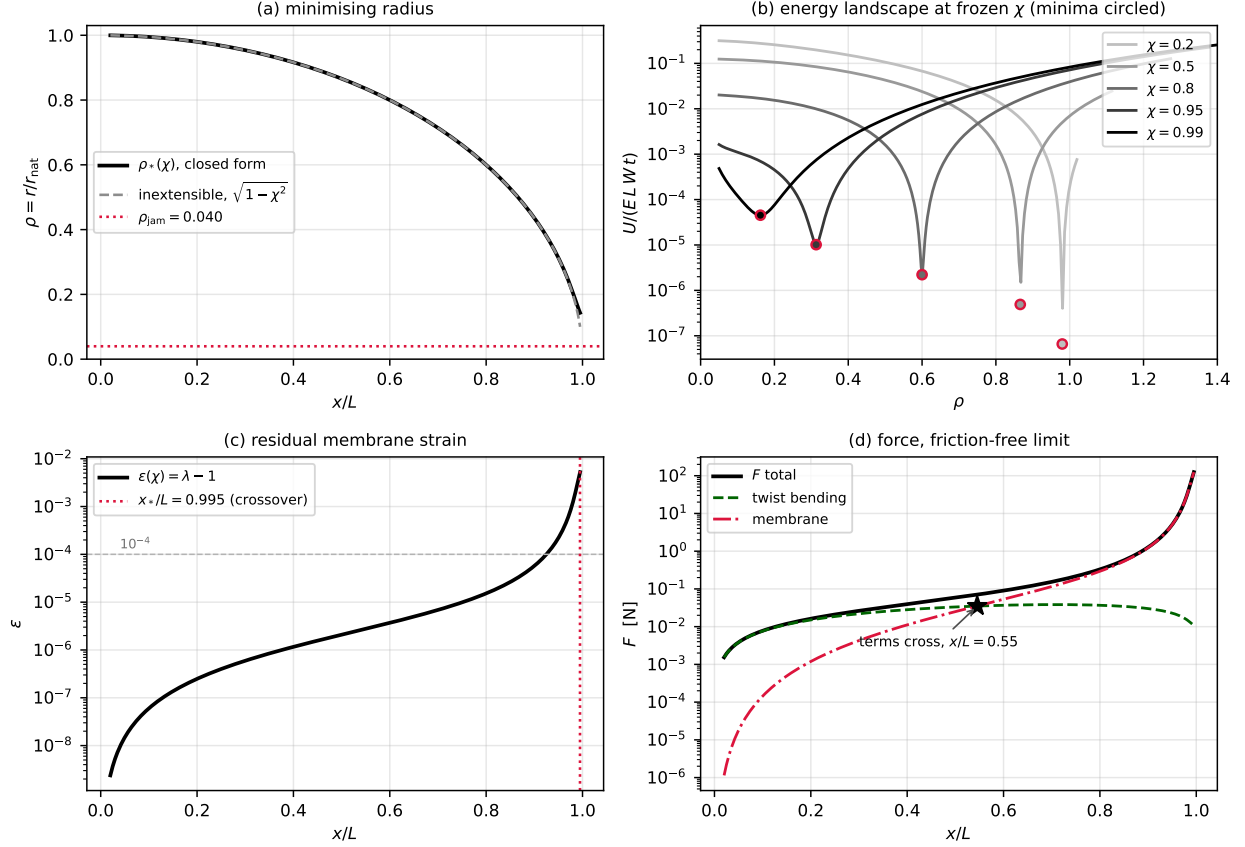


Figure 9: The closed-form solution of Eq. (153) for the coil of Section 9. (a) The minimizing radius, against the inextensible $\sqrt{1-\chi^2}$ and the packing scale ρ_{jam} . (b) The energy at five frozen extensions, showing one interior minimum apiece; the minima are circled. (c) The residual membrane strain, with the crossover x_*/L of Eq. (78) marked. (d) The force and its two terms from Eq. (154), in the friction-free limit $\mathcal{A} = 1$.

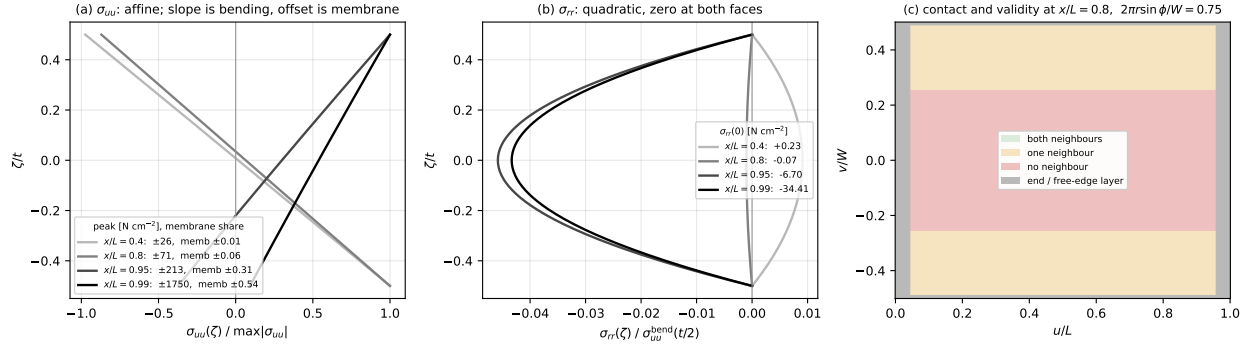


Figure 10: Fields of the minimizing state, and where they are valid. (a) σ_{uu} is affine in ζ by Eq. (117): the offset is membrane, the slope bending, and the legend gives the peak value and the membrane share of it. The slope changes sign between $x/L = 0.8$ and 0.95 . (b) σ_{rr} is the quadratic bump of Eq. (143), normalized by the bending stress at the surface; it vanishes at both faces and changes sign near $x/L \approx 0.7$. (c) The (u, v) plane at $x/L = 0.8$: the boundary layers assumption 10 discards, and the contact gates of Eqs. (40)–(41). Inside the valid region the stress field is uniform in (u, v) by construction, which is why no stress is plotted over it.